```
=== .env dan docker-compose.yml ===

--- .env ---
# ===== Identitas =====
NAMA=muhammad-haikal-fadhil-imani

# ===== PostgreSQL =====
POSTGRES_USER=postgres
POSTGRES_PASSWORD=ifunggul
POSTGRES_DB=kampus_db

# ===== Port eksternal: 22086 dan 44086 =====
PG_PORT=22086
PGADMIN_PORT=44086

# ===== pgAdmin4 =====
PGADMIN_DEFAULT_EMAIL=admin@example.com
PGADMIN_DEFAULT_PASSWORD=pgadmin_086

# ===== Password role aplikasi =====
BACKEND_DEV_PASSWORD=backend_Dev_086
BI_DEV_PASSWORD=bi_Dev_086
DATA_ENGINEER_PASSWORD=data_Engineer_086

--- docker-compose.yml ---
name: proyek1-basisdata

services:
  postgres:
    image: postgres:15
    container_name: postgres_
    restart: unless-stopped
    environment:
      POSTGRES_USER:
      POSTGRES_PASSWORD:
      POSTGRES_DB:
      BACKEND_DEV_PASSWORD:
      BI_DEV_PASSWORD:
      DATA_ENGINEER_PASSWORD:
    ports:
      - "127.0.0.1::5432"
    volumes:
      - pg_data:/var/lib/postgresql/data
      - ./sql:/sql:ro
      - ./tests:/tests:ro
    networks:
      - net_mbd
    healthcheck:
      test: ["CMD-SHELL", "pg_isready -U ${POSTGRES_USER} -d ${POSTGRES_DB}"]
      interval: 5s
      timeout: 5s
      retries: 10

  pgadmin:
    image: dpage/pgadmin4:latest
    container_name: pgadmin_
    restart: unless-stopped
    environment:
      PGADMIN_DEFAULT_EMAIL:
      PGADMIN_DEFAULT_PASSWORD:
    ports:
      - "127.0.0.1::80"
    volumes:
      - pgadmin_data:/var/lib/pgadmin
    networks:
      - net_mbd
    depends_on:
      postgres:
        condition: service_healthy

volumes:
  pg_data:
  pgadmin_data:

networks:
  net_mbd:
    driver: bridge
```


$ docker compose ps
NAME IMAGE COMMAND SERVICE CREATED STATUS PORTS
pgadmin_muhammad-haikal-fadhil-imani dpage/pgadmin4:latest "/entrypoint.sh" pgadmin 4 minutes ago Up 4 minutes 127.0.0.1:44066->80/tcp
postgres_muhammad-haikal-fadhil-imani postgres:15 "docker-entrypoint.s..." postgres 4 minutes ago Up 4 minutes (healthy) 127.0.0.1:32066->5432/t

```
>>> Output 01_setup.sql — Skema SALAM + tabel mahasiswas + 3 role <<<

>>> [1/7] Membuat role (SQL script)
>>> [2/7] Membuat skema SALAM (huruf besar -> wajib kutip ganda)
CREATE SCHEMA
>>> [3/7] Membuat tabel mahasiswas (PK, UNIQUE, CHECK, NOT NULL)
docker : psql:/sql/01_setup.sql:19: NOTICE:  schema "SALAM" already exists, skipping
At line:2 char:13
+ $setupOut = docker exec -e BACKEND_DEV_PASSWORD=backend_Dev_086 -e BI ...
+             ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
    + CategoryInfo          : NotSpecified: (psql:/sql/01_se...xists, skipping:String) [], RemoteE
    + FullyQualifiedErrorId : NativeCommandError

psql:/sql/01_setup.sql:44: NOTICE:  relation "mahasiswas" already exists, skipping
psql:/sql/01_setup.sql:56: NOTICE:  relation "audit_mahasiswas" already exists, skipping
CREATE TABLE
CREATE TABLE
CREATE FUNCTION
CREATE VIEW
CREATE FUNCTION
CREATE FUNCTION
>>> [4/7] Data awal (sebelum trigger audit dipasang)
INSERT 0 0
DROP TRIGGER
CREATE TRIGGER
DROP TRIGGER
CREATE TRIGGER
>>> [5/7] data_engineer dijadikan OWNER semua object di skema SALAM
ALTER SCHEMA
ALTER TABLE
ALTER TABLE
ALTER VIEW
ALTER FUNCTION
ALTER FUNCTION
ALTER FUNCTION
>>> [6/7] GRANT hak akses per role
GRANT
GRANT
GRANT
GRANT
GRANT
ALTER DEFAULT PRIVILEGES
ALTER DEFAULT PRIVILEGES
ALTER DEFAULT PRIVILEGES
>>> [7/7] Verifikasi
                            List of schemas
 Name  |     Owner     |       Access privileges        | Description
-------+---------------+--------------------------------+-------------
 SALAM | data_engineer | data_engineer=UC/data_engineer+|
       |               | backend_dev=U/data_engineer   +|
       |               | bi_dev=U/data_engineer         |
(1 row)

                 List of relations
 Schema |       Name       | Type  |     Owner
--------+------------------+-------+---------------
 SALAM  | audit_mahasiswas | table | data_engineer
 SALAM  | mahasiswas       | table | data_engineer
(2 rows)

            List of roles
  Role name  | Attributes | Member of
-------------+------------+-----------
 backend_dev |            | {}

           List of roles
 Role name | Attributes | Member of
-----------+------------+-----------
 bi_dev    |            | {}

             List of roles
   Role name   | Attributes | Member of
---------------+------------+-----------
 data_engineer |            | {}

>>> SELESAI
```


pgAdmin
Login
Email Address / Username
Password
Forgotten your password?
English
Login

```
=== 01_setup.sql ===
>>> [1/7] Membuat role (SQL script)
>>> [2/7] Membuat skema SALAM (huruf besar -> wajib kutip ganda)
CREATE SCHEMA
>>> [3/7] Membuat tabel mahasiswas (PK, UNIQUE, CHECK, NOT NULL)
CREATE TABLE
CREATE TABLE
CREATE FUNCTION
CREATE VIEW
CREATE FUNCTION
CREATE FUNCTION
>>> [4/7] Data awal (sebelum trigger audit dipasang)
INSERT 0 0
DROP TRIGGER
CREATE TRIGGER
DROP TRIGGER
CREATE TRIGGER
>>> [5/7] data_engineer dijadikan OWNER semua object di skema SALAM
ALTER SCHEMA
ALTER TABLE
ALTER TABLE
ALTER VIEW
ALTER FUNCTION
ALTER FUNCTION
ALTER FUNCTION
>>> [6/7] GRANT hak akses per role
GRANT
GRANT
GRANT
GRANT
GRANT
ALTER DEFAULT PRIVILEGES
ALTER DEFAULT PRIVILEGES
ALTER DEFAULT PRIVILEGES
>>> [7/7] Verifikasi
                            List of schemas
 Name  |     Owner     |       Access privileges        | Description
-------+---------------+--------------------------------+-------------
 SALAM | data_engineer | data_engineer=UC/data_engineer+|
       |               | backend_dev=U/data_engineer   +|
       |               | bi_dev=U/data_engineer         |
(1 row)

                List of relations
 Schema |       Name       | Type  |     Owner
--------+------------------+-------+---------------
 SALAM  | audit_mahasiswas | table | data_engineer
 SALAM  | mahasiswas       | table | data_engineer
(2 rows)

           List of roles
  Role name  | Attributes | Member of
-------------+------------+-----------
 backend_dev |            | {}

           List of roles
 Role name | Attributes | Member of
-----------+------------+-----------
 bi_dev    |            | {}

            List of roles
   Role name    | Attributes | Member of
----------------+------------+-----------
 data_engineer |            | {}

>>> SELESAI
NOTICE: schema "SALAM" already exists, skipping
NOTICE: relation "mahasiswas" already exists, skipping
NOTICE: relation "audit_mahasiswas" already exists, skipping
```

```
=== 01_constraints.sql (semua ERROR + 1 OK) ===

=== 0. Data awal
 id |    nim     |             nama              | ipk  | status
----+------------+-------------------------------+------+--------
  1 | 1247050086 | Muhammad Haikal Fadhil Imani | 3.80 | aktif
  2 | 1247050001 | Sari Wulandari               | 3.45 | aktif
  3 | 1247050002 | Budi Santoso                 | 2.90 | aktif
  4 | 1247050003 | Dewi Lestari                 | 3.10 | cuti
  5 | 1247050004 | Andi Pratama                 | 3.62 | lulus
(5 rows)

=== 1. PRIMARY KEY: id duplikat  [ERROR: pk_mahasiswas]
ERROR: duplicate key value violates unique constraint "pk_mahasiswas"
DETAIL: Key (id)=(1) already exists.

=== 2. UNIQUE: nim duplikat  [ERROR: uq_mahasiswas_nim]
ERROR: duplicate key value violates unique constraint "uq_mahasiswas_nim"
DETAIL: Key (nim)=(1247050086) already exists.

=== 3. UNIQUE: email duplikat  [ERROR: uq_mahasiswas_email]
ERROR: duplicate key value violates unique constraint "uq_mahasiswas_email"
DETAIL: Key (email)=(haikal@example.com) already exists.

=== 4. CHECK: ipk = 4.50 (> 4.00)  [ERROR: ck_mahasiswas_ipk]
ERROR: new row for relation "mahasiswas" violates check constraint "ck_mahasiswas_ipk"

=== 5. CHECK: semester = 0  [ERROR: ck_mahasiswas_semester]
ERROR: new row for relation "mahasiswas" violates check constraint "ck_mahasiswas_semester"

=== 6. CHECK: status tidak valid  [ERROR: ck_mahasiswas_status]
ERROR: new row for relation "mahasiswas" violates check constraint "ck_mahasiswas_status"

=== 7. CHECK: format NIM salah (huruf)  [ERROR: ck_mahasiswas_nim]
ERROR: new row for relation "mahasiswas" violates check constraint "ck_mahasiswas_nim"

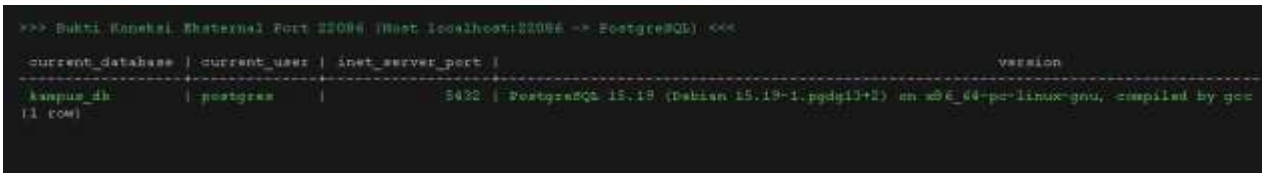
=== 8. CHECK: format email salah  [ERROR: ck_mahasiswas_email]
ERROR: new row for relation "mahasiswas" violates check constraint "ck_mahasiswas_email"

=== 9. NOT NULL: nama NULL  [ERROR: not-null constraint]
ERROR: null value in column "nama" of relation "mahasiswas" violates not-null constraint

=== 10. Data VALID  [OK: 1 baris masuk]
 id |    nim     |      nama       | ipk
----+------------+-----------------+------
 51 | 1247050999 | Mahasiswa Valid | 3.25
(1 row)

INSERT 0 1
=== 11. Ringkasan: hanya data valid yang tersimpan
 id |    nim     |             nama              | ipk  | status
----+------------+-------------------------------+------+--------
  1 | 1247050086 | Muhammad Haikal Fadhil Imani | 3.80 | aktif
  2 | 1247050001 | Sari Wulandari               | 3.45 | aktif
  3 | 1247050002 | Budi Santoso                 | 2.90 | aktif
  4 | 1247050003 | Dewi Lestari                 | 3.10 | cuti
  5 | 1247050004 | Andi Pratama                 | 3.62 | lulus
 51 | 1247050999 | Mahasiswa Valid              | 3.25 | aktif
(6 rows)

=== 12. Bersihkan data uji
DELETE 1
```


>>> Bukti Koneksi Eksternal Port 22086 (Host localhost:22086 -> PostgreSQL) <<<
current_database | current_user | inet_server_port | version
kampus_db | postgres | 5432 | PostgreSQL 15.13 (Debian 15.13-1.pgdg13+1) on x86_64-pc-linux-gnu, compiled by gcc
(1 row)

```
>>> Output 01_constraints.sql — Semua ERROR ditolak, [OK] masuk <<<

=== 0. Data awal
 id |    nim     |             nama              | ipk  | status
----+------------+-------------------------------+------+--------
  1 | 1247050086 | Muhammad Haikal Fadhil Imani | 3.80 | aktif
  2 | 1247050001 | Sari Wulandari               | 3.45 | aktif
  3 | 1247050002 | Budi Santoso                 | 2.90 | aktif
  4 | 1247050003 | Dewi Lestari                 | 3.10 | cuti
  5 | 1247050004 | Andi Pratama                 | 3.62 | lulus
(5 rows)

=== 1. PRIMARY KEY: id duplikat  [ERROR: pk_mahasiswas]
=== 2. UNIQUE: nim duplikat  [ERROR: uq_mahasiswas_nim]
docker : psql:/tests/01_constraints.sql:11: ERROR:  duplicate key value violates unique constraint
At line:2 char:18
+ ... traintOut = docker exec postgres_muhammad-haikal-fadhil-imani psql -U ...
+                 ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
    + CategoryInfo          : NotSpecified: (psql:/tests/01_..."pk_mahasiswas":String) [], RemoteE
    + FullyQualifiedErrorId : NativeCommandError

DETAIL:  Key (id)=(1) already exists.
=== 3. UNIQUE: email duplikat  [ERROR: uq_mahasiswas_email]
psql:/tests/01_constraints.sql:15: ERROR:  duplicate key value violates unique constraint "uq_mahs
DETAIL:  Key (nim)=(1247050086) already exists.
psql:/tests/01_constraints.sql:19: ERROR:  duplicate key value violates unique constraint "uq_mahs
DETAIL:  Key (email)=(haikal@example.com) already exists.
=== 4. CHECK: ipk = 4.50 (> 4.00)  [ERROR: ck_mahasiswas_ipk]
psql:/tests/01_constraints.sql:23: ERROR:  new row for relation "mahasiswas" violates check consti
pk"
DETAIL:  Failing row contains (29, 1247050903, Uji IPK, uji.ipk@example.com, Teknik Informatika, 2
2026-10-31 13:16:43.787662+00, 2026-10-31 13:16:43.787662+00).
=== 5. CHECK: semester = 0  [ERROR: ck_mahasiswas_semester]
=== 6. CHECK: status tidak valid  [ERROR: ck_mahasiswas_status]
psql:/tests/01_constraints.sql:27: ERROR:  new row for relation "mahasiswas" violates check consti
emester"
DETAIL:  Failing row contains (30, 1247050904, Uji Semester, uji.smt@example.com, Teknik Informati
tif, 2026-10-31 13:16:43.787921+00, 2026-10-31 13:16:43.787921+00).
psql:/tests/01_constraints.sql:31: ERROR:  new row for relation "mahasiswas" violates check consti
tatus"
DETAIL:  Failing row contains (31, 1247050905, Uji Status, uji.status@example.com, Teknik Informat
eluar, 2026-10-31 13:16:43.788178+00, 2026-10-31 13:16:43.788178+00).
=== 7. CHECK: format NIM salah (huruf)  [ERROR: ck_mahasiswas_nim]
=== 8. CHECK: format email salah  [ERROR: ck_mahasiswas_email]
psql:/tests/01_constraints.sql:35: ERROR:  new row for relation "mahasiswas" violates check consti
im"
DETAIL:  Failing row contains (32, ABC1234567, Uji Format NIM, uji.fmt@example.com, Teknik Informa
aktif, 2026-10-31 13:16:43.788379+00, 2026-10-31 13:16:43.788379+00).
psql:/tests/01_constraints.sql:39: ERROR:  new row for relation "mahasiswas" violates check consti
mail"
DETAIL:  Failing row contains (33, 1247050906, Uji Email Format, bukan-email, Teknik Informatika,
 2026-10-31 13:16:43.788577+00, 2026-10-31 13:16:43.788577+00).
=== 9. NOT NULL: nama NULL  [ERROR: not-null constraint]
psql:/tests/01_constraints.sql:43: ERROR:  null value in column "nama" of relation "mahasiswas" vi
raint
=== 10. Data VALID  [OK: 1 baris masuk]
DETAIL:  Failing row contains (34, 1247050907, null, uji.null@example.com, Teknik Informatika, 202
26-10-31 13:16:43.788827+00, 2026-10-31 13:16:43.788827+00).
 id |    nim     |       nama       | ipk
----+------------+------------------+------
 35 | 1247050999 | Mahasiswa Valid | 3.25
(1 row)

INSERT 0 1
=== 11. Ringkasan: hanya data valid yang tersimpan
 id |    nim     |             nama              | ipk  | status
----+------------+-------------------------------+------+--------
  1 | 1247050086 | Muhammad Haikal Fadhil Imani | 3.80 | aktif
  2 | 1247050001 | Sari Wulandari               | 3.45 | aktif
  3 | 1247050002 | Budi Santoso                 | 2.90 | aktif
  4 | 1247050003 | Dewi Lestari                 | 3.10 | cuti
  5 | 1247050004 | Andi Pratama                 | 3.62 | lulus
 35 | 1247050999 | Mahasiswa Valid              | 3.25 | aktif
(6 rows)

=== 12. Bersihkan data uji
DELETE 1
```

```
=== 02_backend.sql (backend_dev) ===

=== Login sebagai:
 current_user | session_user
--------------+--------------
 backend_dev  | backend_dev
(1 row)

=== [OK] CREATE (INSERT)
 id |    nim     |     nama     | ipk
----+------------+--------------+------
 52 | 1247050777 | Backend Test | 3.00
(1 row)

INSERT 0 1
=== [OK] READ (SELECT)
 id |    nim     |     nama     | ipk
----+------------+--------------+------
 52 | 1247050777 | Backend Test | 3.00
(1 row)

=== [OK] UPDATE
    nim     | ipk  |          updated_at
------------+------+-------------------------------
 1247050777 | 3.75 | 2026-10-31 13:17:16.019936+00
(1 row)

UPDATE 1
=== [OK] Transaksi: UPDATE lalu ROLLBACK (semester kembali semula)
BEGIN
UPDATE 1
    nim     | semester_dalam_transaksi
------------+--------------------------
 1247050777 |                        6
(1 row)

ROLLBACK
    nim     | semester_setelah_rollback
------------+---------------------------
 1247050777 |                         5
(1 row)

=== [OK] DELETE
DELETE 1
=== [OK] Jejak audit otomatis (db_user = backend_dev)
 audit_id |             waktu             |   db_user   |  aksi  |    nim
----------+-------------------------------+-------------+--------+------------
       24 | 2026-10-31 13:17:16.02321+00  | backend_dev | DELETE | 1247050777
       22 | 2026-10-31 13:17:16.019936+00 | backend_dev | UPDATE | 1247050777
       21 | 2026-10-31 13:17:16.015129+00 | backend_dev | INSERT | 1247050777
       20 | 2026-10-31 13:17:08.730972+00 | postgres    | DELETE | 1247050999
(4 rows)

=== [ERROR] CREATE TABLE -> permission denied for schema SALAM
ERROR: permission denied for schema SALAM

=== [ERROR] ALTER TABLE -> must be owner
ERROR: must be owner of table mahasiswas

=== [ERROR] DROP TABLE -> must be owner
ERROR: must be owner of table mahasiswas
```

```
>>> Output 02_backend.sql — Role backend_dev (CRUD OK, DDL ERROR) <<<

=== Login sebagai:
 current_user | session_user
--------------+--------------
 backend_dev  | backend_dev
(1 row)

=== [OK] CREATE (INSERT)
 id |    nim     |     nama     | ipk
----+------------+--------------+------
 36 | 1247050777 | Backend Test | 3.00
(1 row)

INSERT 0 1
=== [OK] READ (SELECT)
 id |    nim     |     nama     | ipk
----+------------+--------------+------
 36 | 1247050777 | Backend Test | 3.00
(1 row)

=== [OK] UPDATE
    nim     | ipk  |          updated_at
------------+------+-------------------------------
 1247050777 | 3.75 | 2026-10-31 13:16:54.414965+00
(1 row)

UPDATE 1
=== [OK] Transaksi: UPDATE lalu ROLLBACK (semester kembali semula)
BEGIN
UPDATE 1
    nim     | semester_dalam_transaksi
------------+--------------------------
 1247050777 |                        6
(1 row)

ROLLBACK
    nim     | semester_setelah_rollback
------------+---------------------------
 1247050777 |                         5
(1 row)

=== [OK] DELETE
DELETE 1
=== [OK] Jejak audit otomatis (db_user = backend_dev)
 audit_id |             waktu             |   db_user    |  aksi  |    nim
----------+-------------------------------+--------------+--------+------------
       15 | 2026-10-31 13:16:54.418536+00 | backend_dev  | DELETE | 1247050777
       13 | 2026-10-31 13:16:54.414965+00 | backend_dev  | UPDATE | 1247050777
       12 | 2026-10-31 13:16:54.410212+00 | backend_dev  | INSERT | 1247050777
       11 | 2026-10-31 13:16:43.792446+00 | postgres     | DELETE | 1247050999
(4 rows)

=== [ERROR] CREATE TABLE -> permission denied for schema SALAM
=== [ERROR] ALTER TABLE -> must be owner
docker : psql:/tests/02_backend.sql:34: ERROR:  permission denied for schema SALAM
At line:2 char:15
+ ... ackendOut = docker exec postgres_muhammad-haikal-fadhil-imani sh -c " ...
+                 ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
    + CategoryInfo          : NotSpecified: (psql:/tests/02_...or schema SALAM:String) [], RemoteE
    + FullyQualifiedErrorId : NativeCommandError

LINE 1: CREATE TABLE "SALAM".coba_backend (id INT);
                     ^
psql:/tests/02_backend.sql:37: ERROR:  must be owner of table mahasiswas
=== [ERROR] DROP TABLE -> must be owner
psql:/tests/02_backend.sql:40: ERROR:  must be owner of table mahasiswas
```

```
=== 03_bi.sql (bi_dev) ===

=== Login sebagai:
 current_user | session_user
--------------+--------------
 bi_dev       | bi_dev
(1 row)

=== [OK] SELECT tabel
 id |    nim     |             nama              | ipk  | status
----+------------+-------------------------------+------+--------
  1 | 1247050086 | Muhammad Haikal Fadhil Imani | 3.80 | aktif
  2 | 1247050001 | Sari Wulandari               | 3.45 | aktif
  3 | 1247050002 | Budi Santoso                 | 2.90 | aktif
  4 | 1247050003 | Dewi Lestari                 | 3.10 | cuti
  5 | 1247050004 | Andi Pratama                 | 3.62 | lulus
(5 rows)

=== [OK] SELECT view (dengan predikat IPK)
    nim     |             nama              | ipk  |     predikat     | status
------------+-------------------------------+------+------------------+--------
 1247050086 | Muhammad Haikal Fadhil Imani | 3.80 | Dengan Pujian    | aktif
 1247050004 | Andi Pratama                 | 3.62 | Dengan Pujian    | lulus
 1247050001 | Sari Wulandari               | 3.45 | Sangat Memuaskan | aktif
 1247050003 | Dewi Lestari                 | 3.10 | Sangat Memuaskan | cuti
 1247050002 | Budi Santoso                 | 2.90 | Memuaskan        | aktif
(5 rows)

=== [OK] Agregasi untuk BI
     predikat     | jumlah | rata_ipk
------------------+--------+----------
 Dengan Pujian    |      2 |     3.71
 Sangat Memuaskan |      2 |     3.28
 Memuaskan        |      1 |     2.90
(3 rows)

=== [ERROR] INSERT -> permission denied
ERROR: permission denied for table mahasiswas

=== [ERROR] UPDATE -> permission denied
ERROR: permission denied for table mahasiswas

=== [ERROR] DELETE -> permission denied
ERROR: permission denied for table mahasiswas

=== [ERROR] CREATE TABLE -> permission denied for schema SALAM
ERROR: permission denied for schema SALAM

=== [ERROR] DROP VIEW -> must be owner
ERROR: must be owner of view v_mahasiswa_ringkas
```

```
>>> Output 03_bi.sql — Role bi_dev (SELECT OK, Write/DDL ERROR) <<<

=== Login sebagai:
 current_user | session_user
--------------+--------------
 bi_dev       | bi_dev
(1 row)

=== [OK] SELECT tabel
 id |    nim     |             nama              | ipk  | status
----+------------+-------------------------------+------+--------
  1 | 1247050086 | Muhammad Haikal Fadhil Imani | 3.80 | aktif
  2 | 1247050001 | Sari Wulandari               | 3.45 | aktif
  3 | 1247050002 | Budi Santoso                 | 2.90 | aktif
  4 | 1247050003 | Dewi Lestari                 | 3.10 | cuti
  5 | 1247050004 | Andi Pratama                 | 3.62 | lulus
(5 rows)

=== [OK] SELECT view (dengan predikat IPK)
    nim     |             nama              | ipk  |     predikat     | status
------------+-------------------------------+------+------------------+--------
 1247050086 | Muhammad Haikal Fadhil Imani | 3.80 | Dengan Pujian    | aktif
 1247050004 | Andi Pratama                 | 3.62 | Dengan Pujian    | lulus
 1247050001 | Sari Wulandari               | 3.45 | Sangat Memuaskan | aktif
 1247050003 | Dewi Lestari                 | 3.10 | Sangat Memuaskan | cuti
 1247050002 | Budi Santoso                 | 2.90 | Memuaskan        | aktif
(5 rows)

=== [OK] Agregasi untuk BI
     predikat     | jumlah | rata_ipk
------------------+--------+----------
 Dengan Pujian    |      2 |     3.71
 Sangat Memuaskan |      2 |     3.28
 Memuaskan        |      1 |     2.90
(3 rows)

=== [ERROR] INSERT -> permission denied
=== [ERROR] UPDATE -> permission denied
=== [ERROR] DELETE -> permission denied
docker : psql:/tests/03_bi.sql:19: ERROR:  permission denied for table mahasiswas
At line:4 char:10
+ $biOut = docker exec postgres_muhammad-haikal-fadhil-imani sh -c "PGP ...
+          ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
    + CategoryInfo          : NotSpecified: (psql:/tests/03_...able mahasiswas:String) [], RemoteE
    + FullyQualifiedErrorId : NativeCommandError

psql:/tests/03_bi.sql:22: ERROR:  permission denied for table mahasiswas
psql:/tests/03_bi.sql:25: ERROR:  permission denied for table mahasiswas
=== [ERROR] CREATE TABLE -> permission denied for schema SALAM
psql:/tests/03_bi.sql:28: ERROR:  permission denied for schema SALAM
LINE 1: CREATE TABLE "SALAM".coba_bi (id INT);
                     ^
=== [ERROR] DROP VIEW -> must be owner
psql:/tests/03_bi.sql:31: ERROR:  must be owner of view v_mahasiswa_ringkas
```

```
=== 04_engineer.sql (data_engineer) ===
=== Login sebagai:
 current_user  | session_user
---------------+--------------
 data_engineer | data_engineer
(1 row)

=== [OK] CREATE TABLE
CREATE TABLE
=== [OK] MODIFY: ALTER TABLE ADD COLUMN
ALTER TABLE
                    Table "SALAM.krs_test"
 Column |         Type          | Collation | Nullable | Default
--------+-----------------------+-----------+----------+---------
 id     | integer               |           | not null |
 nim    | character varying(10) |           |          |
 matkul | character varying(50) |           |          |
 sks    | smallint              |           |          | 3
Indexes:
    "krs_test_pkey" PRIMARY KEY, btree (id)
Foreign-key constraints:
    "krs_test_nim_fkey" FOREIGN KEY (nim) REFERENCES "SALAM".mahasiswas(nim)

=== [OK] CRUD pada tabel baru
INSERT 0 1
 id |    nim     |        matkul        | sks
----+------------+----------------------+-----
  1 | 1247050086 | Manajemen Basis Data |   3
(1 row)

UPDATE 1
DELETE 1
=== [OK] CRUD pada tabel mahasiswas
INSERT 0 1
UPDATE 1
    nim     |     nama      | ipk
------------+---------------+------
 1247050666 | Engineer Test | 3.40
(1 row)

DELETE 1
=== [OK] CREATE VIEW, lalu MODIFY (ALTER) tabel yang sudah ada
CREATE VIEW
ALTER TABLE
ALTER TABLE
=== [OK] DROP object
DROP VIEW
DROP TABLE
=== Verifikasi: object uji sudah hilang, tabel utama tetap ada
                List of relations
 Schema |       Name       | Type  |     Owner
--------+------------------+-------+---------------
 SALAM  | audit_mahasiswas | table | data_engineer
 SALAM  | mahasiswas       | table | data_engineer
(2 rows)
```

```
>>> Output 04_engineer.sql — Role data_engineer (DDL + CRUD semua OK) <<<

=== Login sebagai:
 current_user  | session_user
---------------+--------------
 data_engineer | data_engineer
(1 row)

=== [OK] CREATE TABLE
CREATE TABLE
=== [OK] MODIFY: ALTER TABLE ADD COLUMN
ALTER TABLE
                    Table "SALAM.krs_test"
 Column |          Type          | Collation | Nullable | Default
--------+------------------------+-----------+----------+---------
 id     | integer                |           | not null |
 nim    | character varying(10)  |           |          |
 matkul | character varying(50)  |           |          |
 sks    | smallint               |           |          | 3
Indexes:
    "krs_test_pkey" PRIMARY KEY, btree (id)
Foreign-key constraints:
    "krs_test_nim_fkey" FOREIGN KEY (nim) REFERENCES "SALAM".mahasiswas(nim)

=== [OK] CRUD pada tabel baru
INSERT 0 1
 id |    nim     |        matkul         | sks
----+------------+-----------------------+-----
  1 | 1247050086 | Manajemen Basis Data |   3
(1 row)

UPDATE 1
DELETE 1
=== [OK] CRUD pada tabel mahasiswas
INSERT 0 1
UPDATE 1
    nim     |     nama      | ipk
------------+---------------+------
 1247050666 | Engineer Test | 3.40
(1 row)

DELETE 1
=== [OK] CREATE VIEW, lalu MODIFY (ALTER) tabel yang sudah ada
CREATE VIEW
ALTER TABLE
ALTER TABLE
=== [OK] DROP object
DROP VIEW
DROP TABLE
=== Verifikasi: object uji sudah hilang, tabel utama tetap ada
                List of relations
 Schema |       Name       | Type  |     Owner
--------+------------------+-------+---------------
 SALAM  | audit_mahasiswas | table | data_engineer
 SALAM  | mahasiswas       | table | data_engineer
(2 rows)
```

```
>>> Tabel audit_mahasiswas — Jejak INSERT/UPDATE/DELETE tiap role <<<
```

audit_id	waktu	db_user	aksi	nim
1	2026-10-31 13:12:50.170956+00	postgres	INSERT	1247050999
2	2026-10-31 13:12:50.17484+00	postgres	DELETE	1247050999
3	2026-10-31 13:13:00.490232+00	backend_dev	INSERT	1247050777
4	2026-10-31 13:13:00.494653+00	backend_dev	UPDATE	1247050777
6	2026-10-31 13:13:00.497737+00	backend_dev	DELETE	1247050777
7	2026-10-31 13:13:14.264187+00	data_engineer	INSERT	1247050666
8	2026-10-31 13:13:14.267343+00	data_engineer	UPDATE	1247050666
9	2026-10-31 13:13:14.269205+00	data_engineer	DELETE	1247050666
10	2026-10-31 13:16:43.789033+00	postgres	INSERT	1247050999
11	2026-10-31 13:16:43.792446+00	postgres	DELETE	1247050999
12	2026-10-31 13:16:54.410212+00	backend_dev	INSERT	1247050777
13	2026-10-31 13:16:54.414965+00	backend_dev	UPDATE	1247050777
15	2026-10-31 13:16:54.418536+00	backend_dev	DELETE	1247050777
16	2026-10-31 13:16:54.921562+00	data_engineer	INSERT	1247050666
17	2026-10-31 13:16:54.925127+00	data_engineer	UPDATE	1247050666
18	2026-10-31 13:16:54.927549+00	data_engineer	DELETE	1247050666

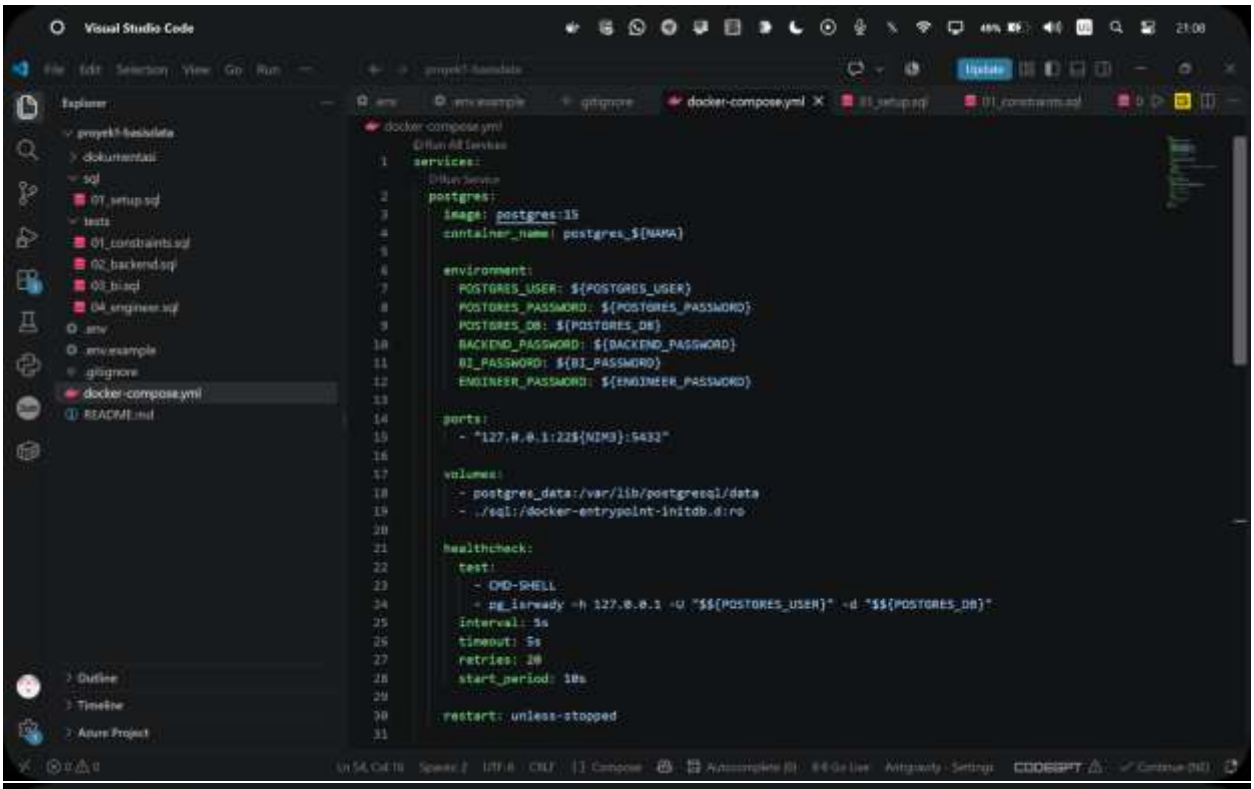
(16 rows)

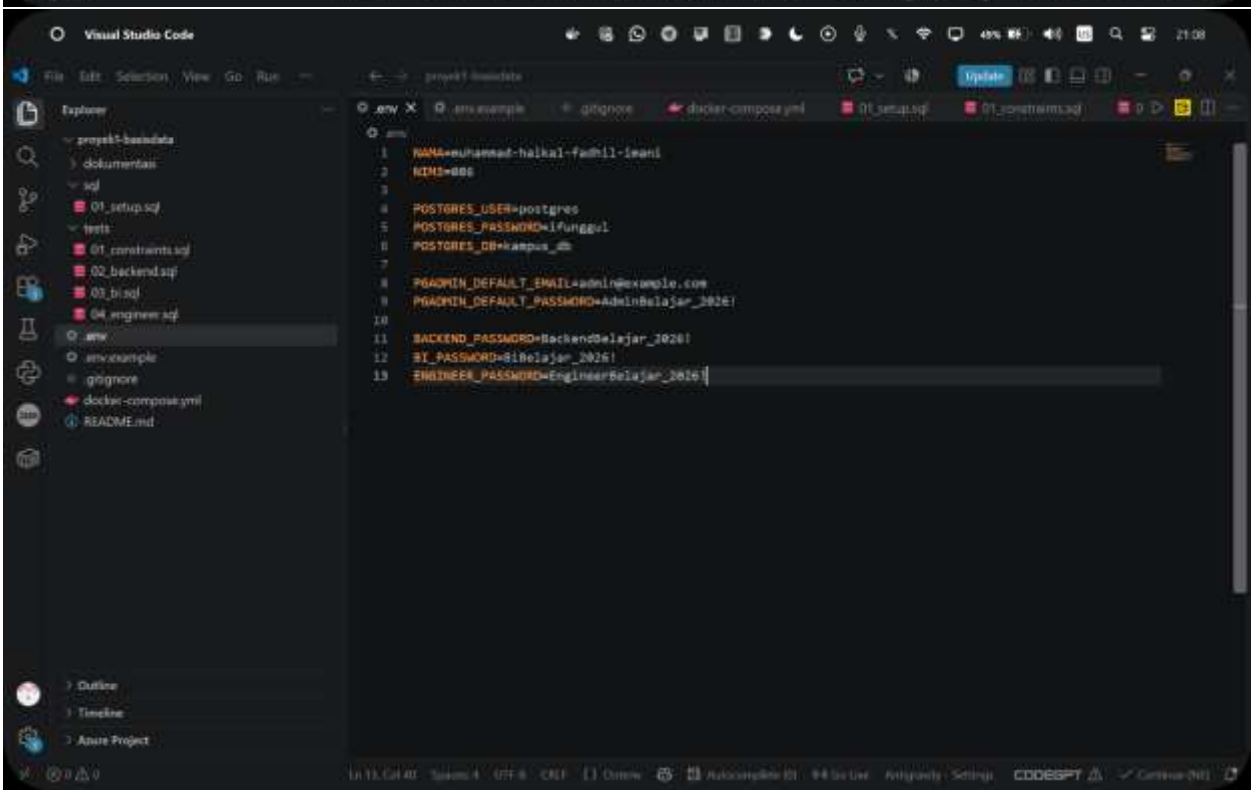
```
=== audit_mahasiswas (SELECT) ===
```

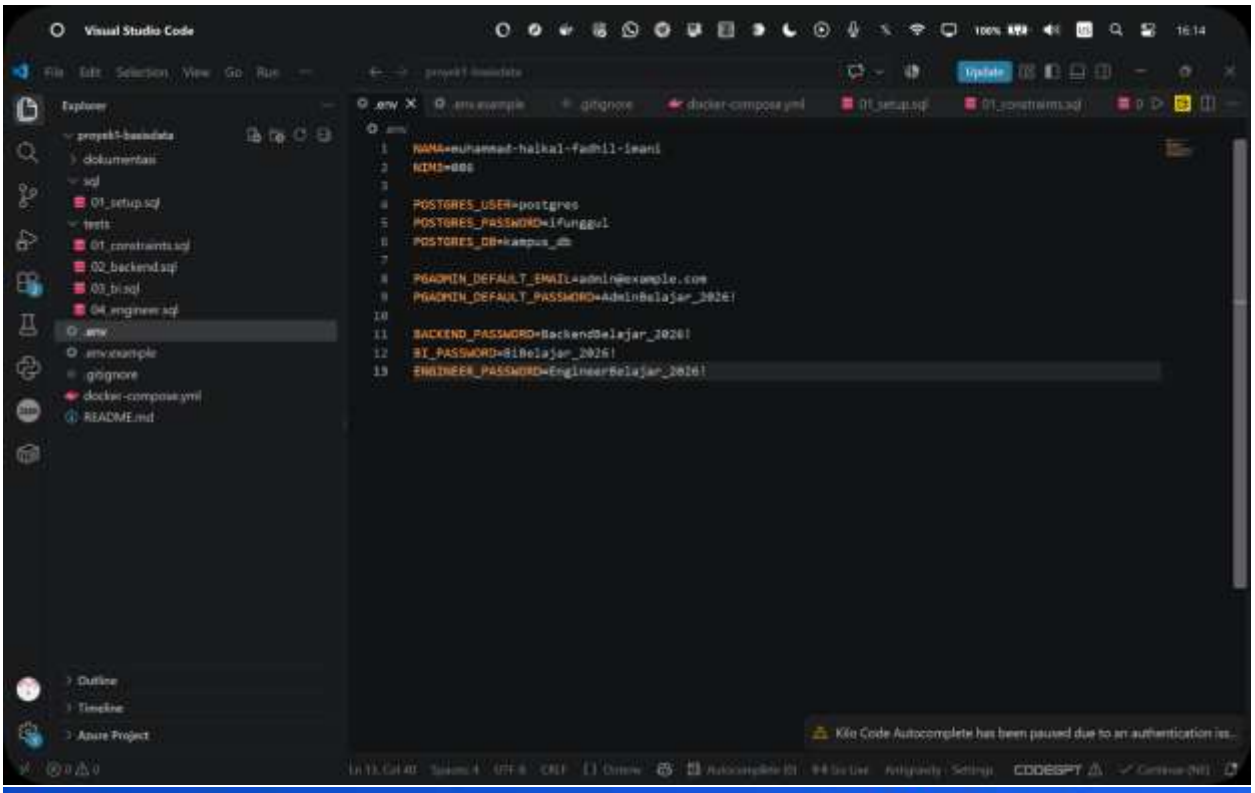
```
SELECT audit_id, waktu, db_user, aksi, nim FROM "SALAM".audit_mahasiswas ORDER BY audit_id;
```

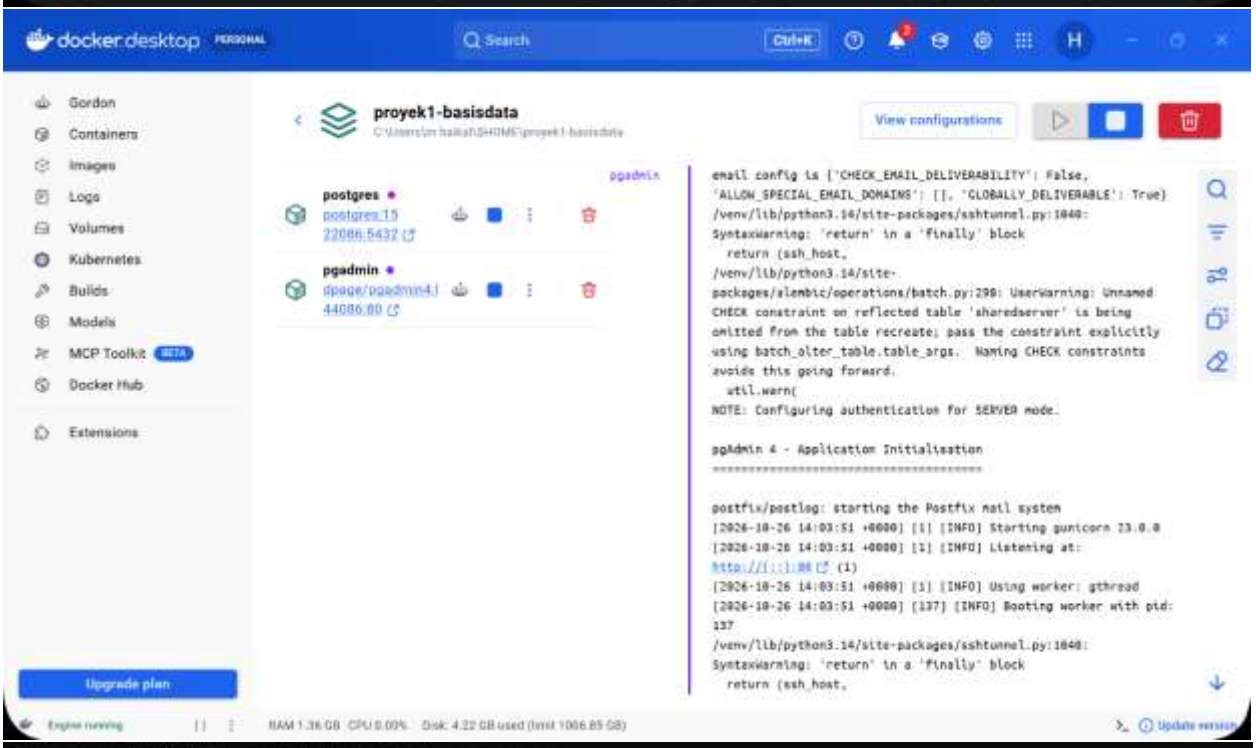
audit_id	waktu	db_user	aksi	nim
1	2026-10-31 13:12:50.170956+00	postgres	INSERT	1247050999
2	2026-10-31 13:12:50.17484+00	postgres	DELETE	1247050999
3	2026-10-31 13:13:00.490232+00	backend_dev	INSERT	1247050777
4	2026-10-31 13:13:00.494653+00	backend_dev	UPDATE	1247050777
6	2026-10-31 13:13:00.497737+00	backend_dev	DELETE	1247050777
7	2026-10-31 13:13:14.264187+00	data_engineer	INSERT	1247050666
8	2026-10-31 13:13:14.267343+00	data_engineer	UPDATE	1247050666
9	2026-10-31 13:13:14.269205+00	data_engineer	DELETE	1247050666
10	2026-10-31 13:16:43.789033+00	postgres	INSERT	1247050999
11	2026-10-31 13:16:43.792446+00	postgres	DELETE	1247050999
12	2026-10-31 13:16:54.410212+00	backend_dev	INSERT	1247050777
13	2026-10-31 13:16:54.414965+00	backend_dev	UPDATE	1247050777
15	2026-10-31 13:16:54.418536+00	backend_dev	DELETE	1247050777
16	2026-10-31 13:16:54.921562+00	data_engineer	INSERT	1247050666
17	2026-10-31 13:16:54.925127+00	data_engineer	UPDATE	1247050666
18	2026-10-31 13:16:54.927549+00	data_engineer	DELETE	1247050666
19	2026-10-31 13:17:08.67557+00	postgres	INSERT	1247050999
20	2026-10-31 13:17:08.730972+00	postgres	DELETE	1247050999
21	2026-10-31 13:17:16.015129+00	backend_dev	INSERT	1247050777
22	2026-10-31 13:17:16.019936+00	backend_dev	UPDATE	1247050777
24	2026-10-31 13:17:16.02321+00	backend_dev	DELETE	1247050777
25	2026-10-31 13:17:34.239592+00	data_engineer	INSERT	1247050666
26	2026-10-31 13:17:34.242623+00	data_engineer	UPDATE	1247050666
27	2026-10-31 13:17:34.244942+00	data_engineer	DELETE	1247050666

(24 rows)

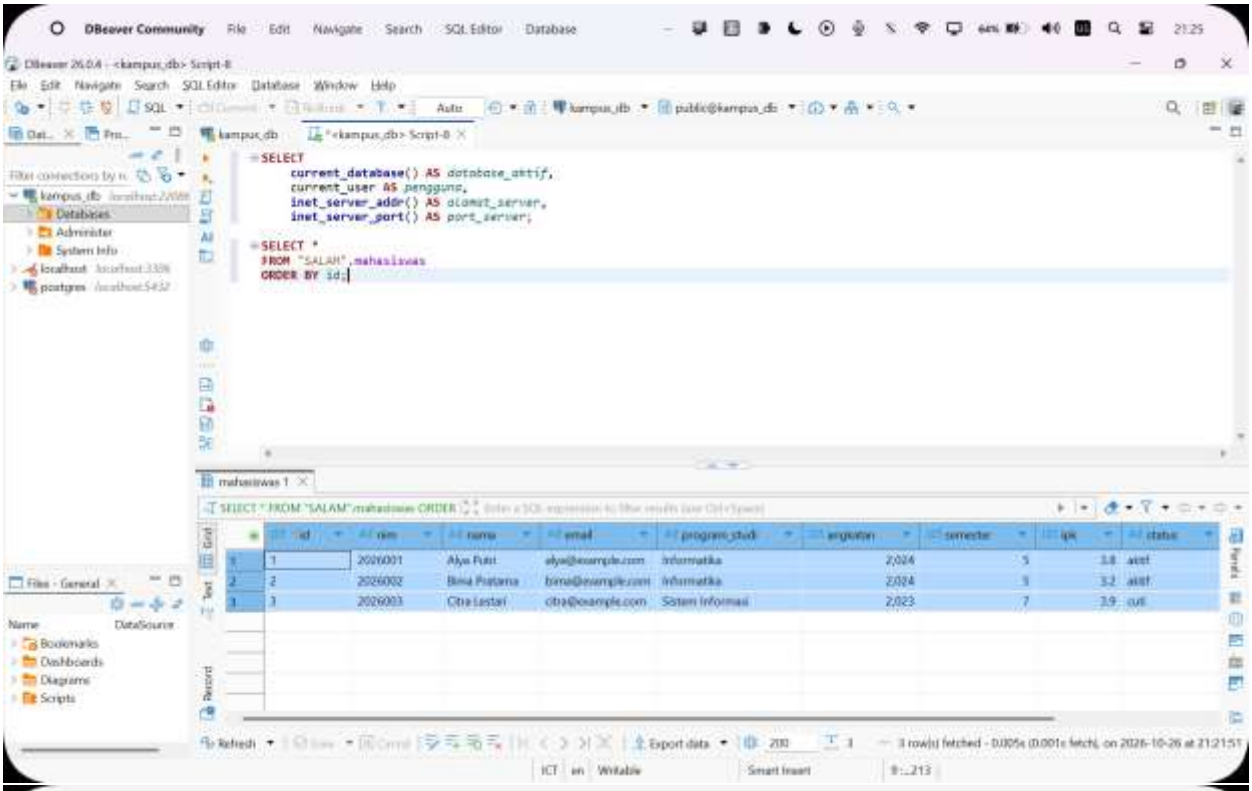

Visual Studio Code
docker-compose.yml
services:
postgres:
image: postgres:15
container_name: postgres_${NAMA}
environment:
POSTGRES_USER: ${POSTGRES_USER}
POSTGRES_PASSWORD: ${POSTGRES_PASSWORD}
POSTGRES_DB: ${POSTGRES_DB}
BACKEND_PASSWORD: ${BACKEND_PASSWORD}
BI_PASSWORD: ${BI_PASSWORD}
ENGINEER_PASSWORD: ${ENGINEER_PASSWORD}
ports:
- "127.0.0.1:22${NIM3}:5432"
volumes:
- postgres_data:/var/lib/postgresql/data
- ./sql:/docker-entrypoint-initdb.d:ro
healthcheck:
test:
- CMD-SHELL
- pg_isready -h 127.0.0.1 -U "$${POSTGRES_USER}" -d "$${POSTGRES_DB}"
interval: 5s
timeout: 5s
retries: 20
start_period: 10s
restart: unless-stopped

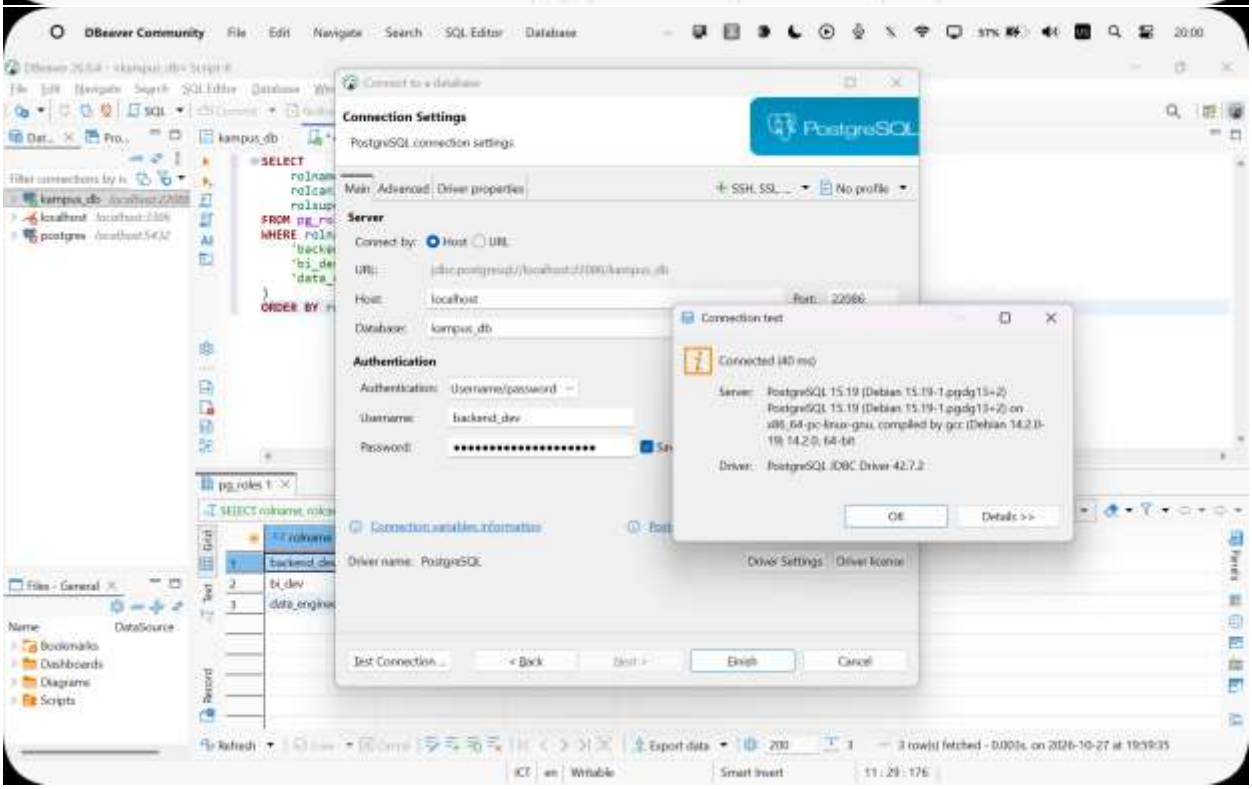

Visual Studio Code
.env
NAMA=muhammad-haikal-fadhil-imani
NIM3=006
POSTGRES_USER=postgres
POSTGRES_PASSWORD=ifunggul
POSTGRES_DB=kampus_db
PGADMIN_DEFAULT_EMAIL=admin@example.com
PGADMIN_DEFAULT_PASSWORD=AdminBelajar_2026!
BACKEND_PASSWORD=BackendBelajar_2026!
BI_PASSWORD=BiBelajar_2026!
ENGINEER_PASSWORD=EngineerBelajar_2026!

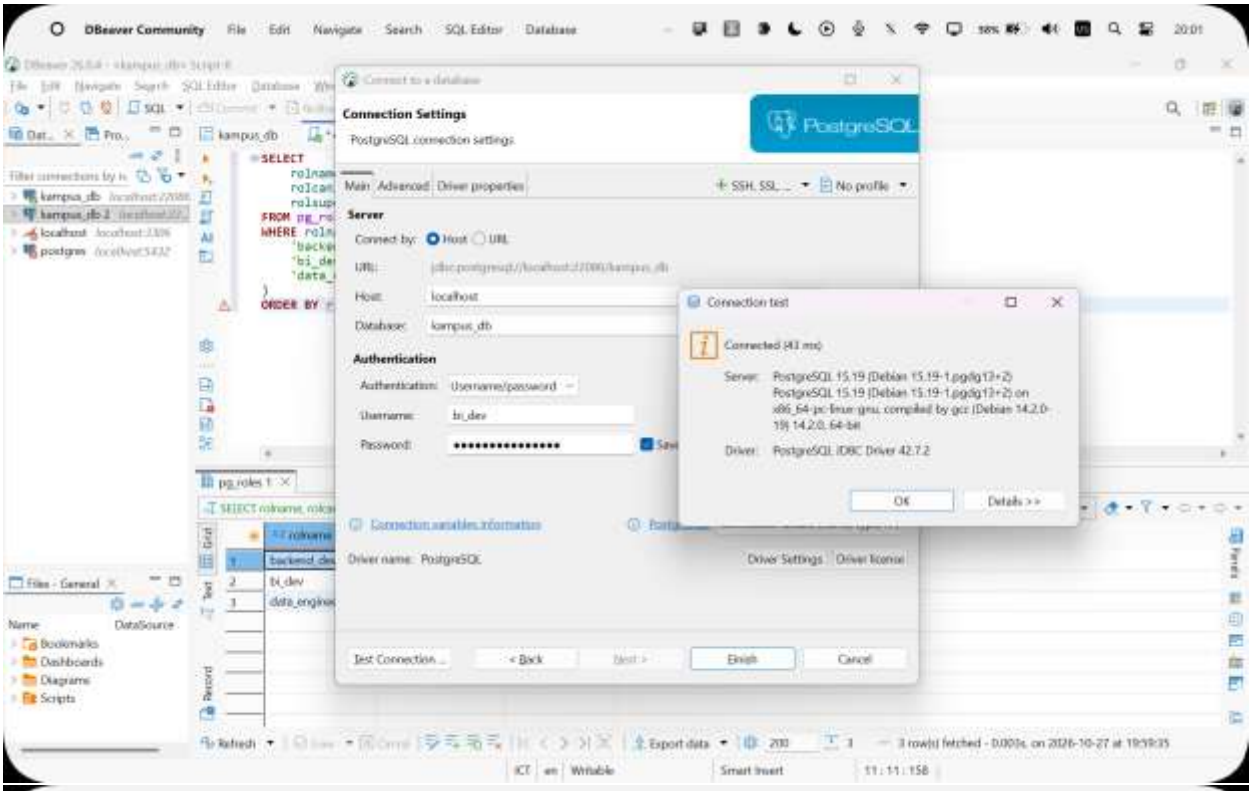


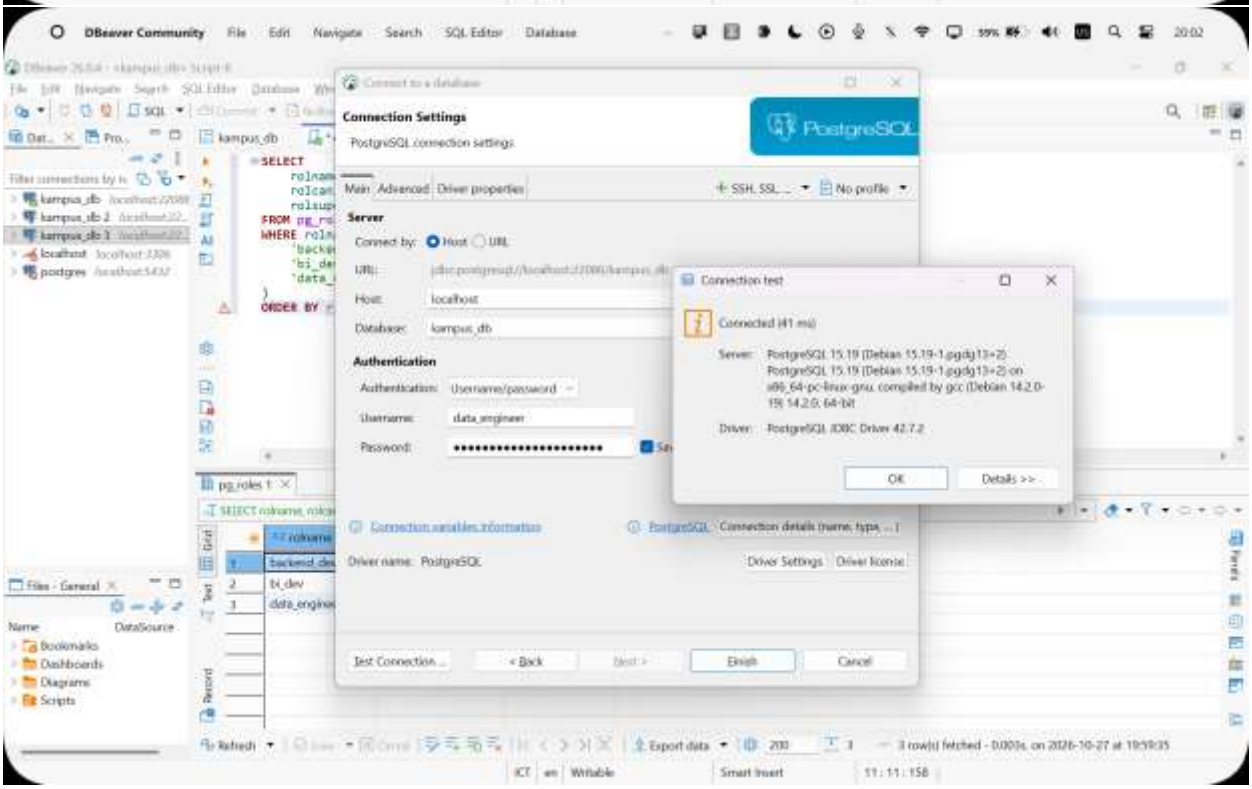


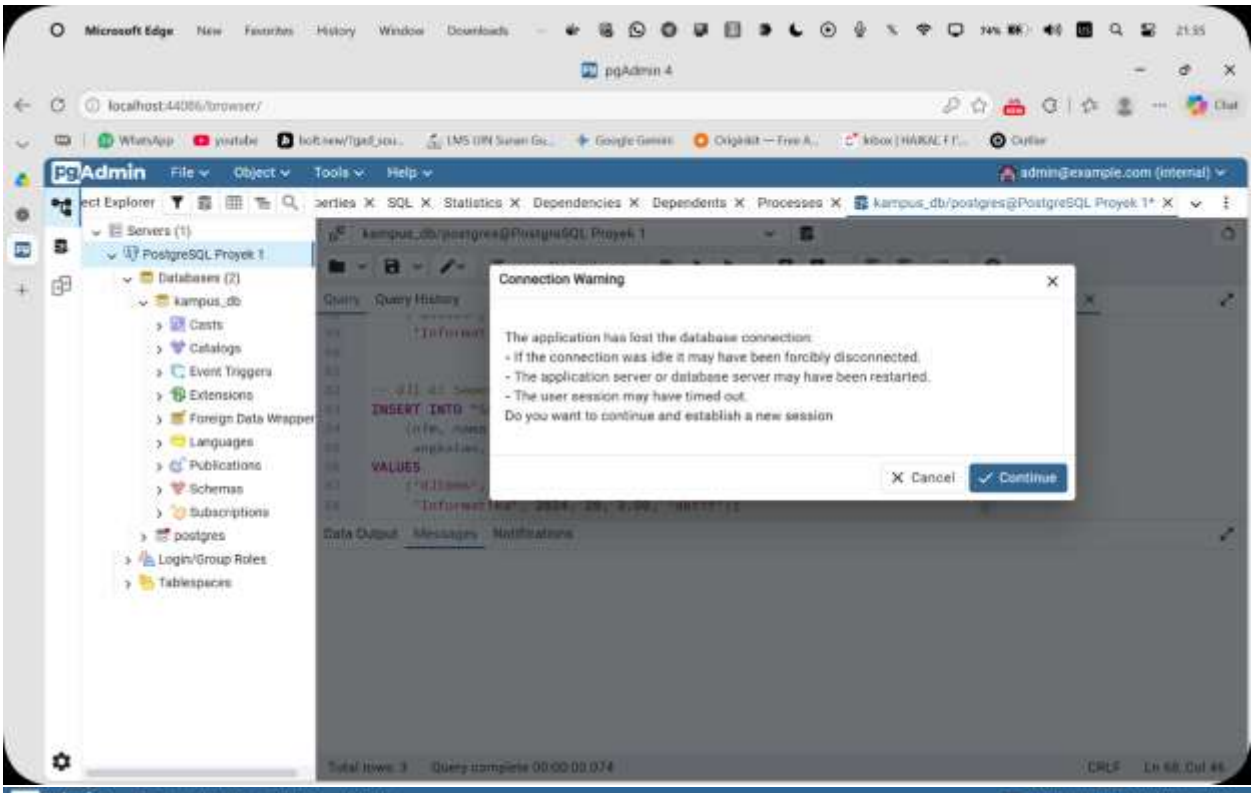

```
C:\Users\m haikal\$HOME\proyek1-basisdata>docker compose ps
NAME                                    IMAGE                COMMAND                  SERVICE    CREATED          STATUS
PORTS
pgadmin_muhammad-haikal-fadhil-imani    dpage/pgadmin4:latest   "/entrypoint.sh"         pgadmin    47 seconds ago   Up 40 seconds
         127.0.0.1:44086->80/tcp
postgres_muhammad-haikal-fadhil-imani   postgres:15          "docker-entrypoint.s…"   postgres   47 seconds ago   Up 45 seconds (h
ealthy)   127.0.0.1:22086->5432/tcp
```

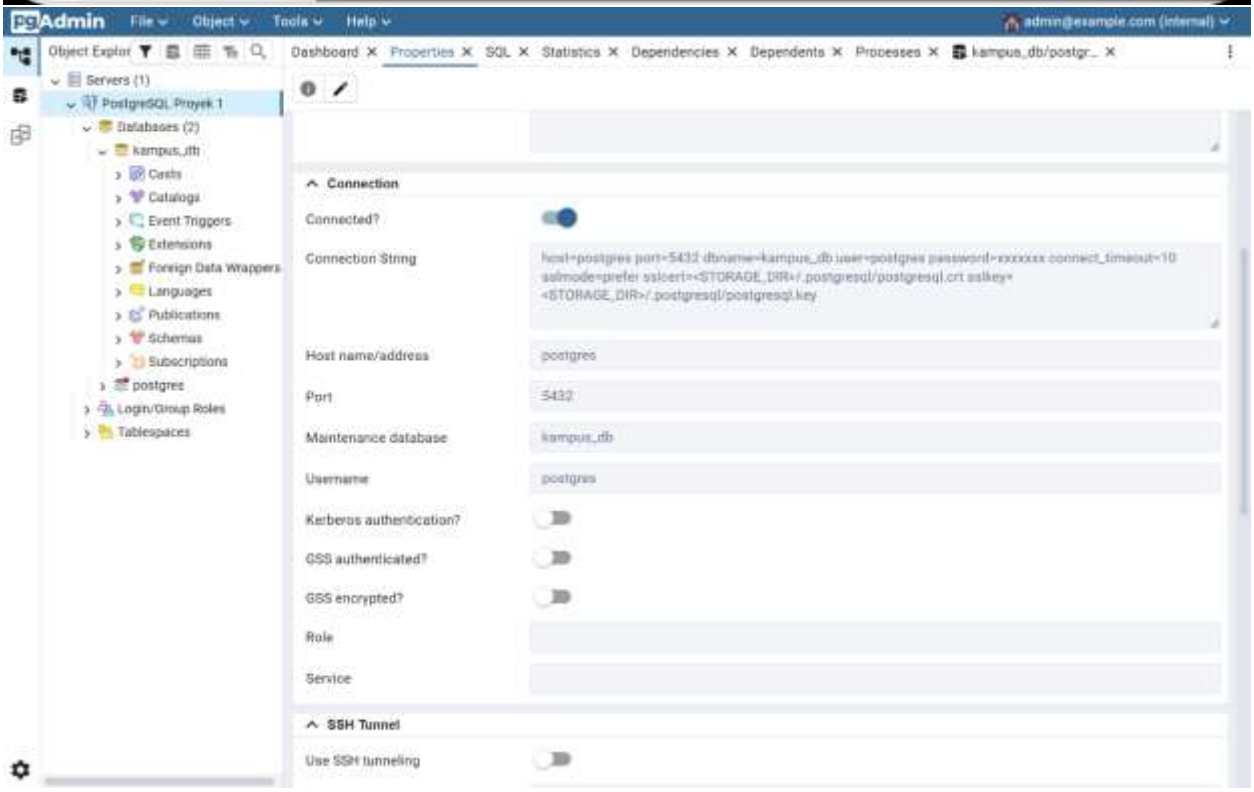

DBeaver Community
File Edit Navigate Search SQL Editor Database
kampus_db
Script-8
SELECT
current_database() AS database_aktif,
current_user AS pengguna,
inet_server_addr() AS alamat_server,
inet_server_port() AS port_server;
SELECT *
FROM "SALAH".mahasiswas
ORDER BY id;
mahasiswas 1
id nim nama email program_studi angkatan semester ipk status
1 2026001 Alya Putri alya@example.com Informatika 2024 5 3.8 aktif
2 2026002 Bima Pratama bima@example.com Informatika 2024 5 3.2 aktif
3 2026003 Citra Lestari citra@example.com Sistem Informasi 2023 7 2.9 cuti
Refresh
Export data
3 row(s) fetched - 0.005s (0.001s fetch), on 2026-10-26 at 21:21:51

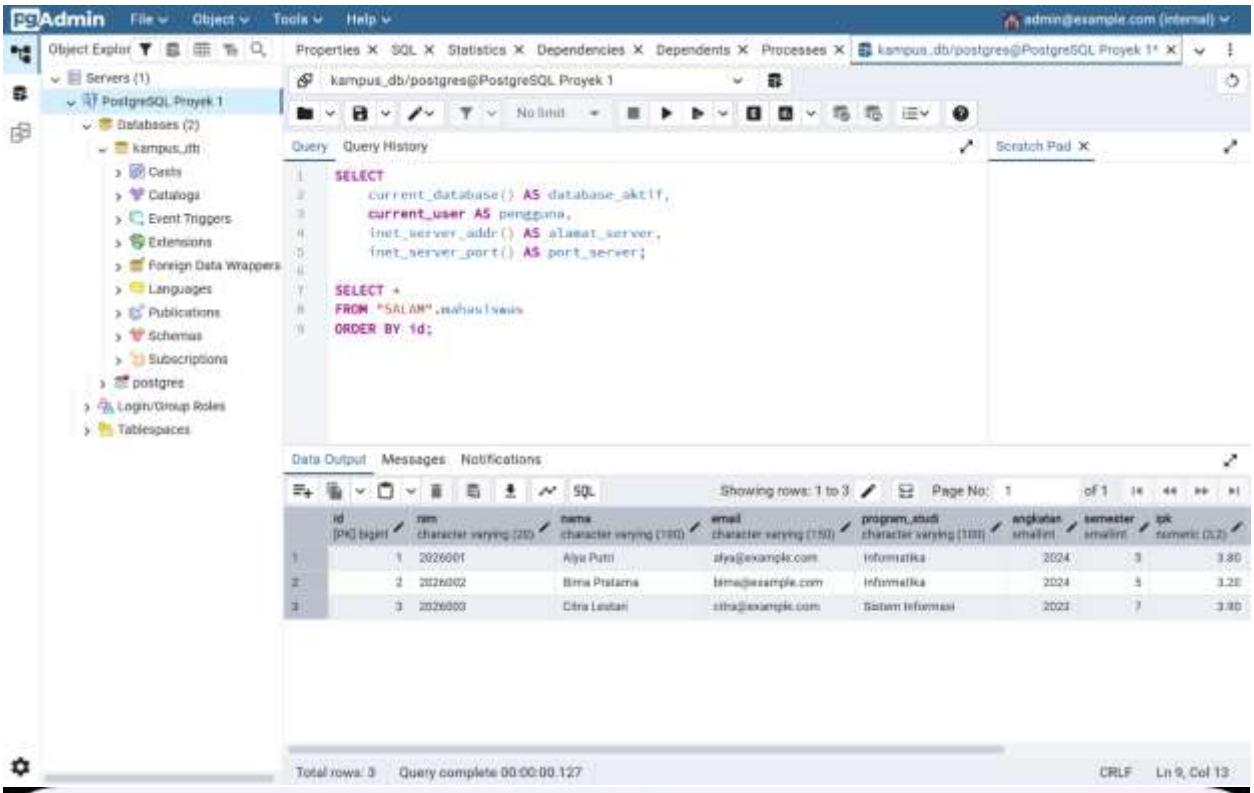

DBeaver Community
Connect to a database
Connection Settings
PostgreSQL connection settings
PostgreSQL
Main Advanced Driver properties
SSH, SSL
No profile
Server
Connect by: Host URL
URL:
Host: localhost
Port: 22086
Database: kampus_db
Authentication
Authentication: Username/password
Username: backend_dev
Password:
Connection variables information
Driver name: PostgreSQL
Driver Settings
Driver license
Test Connection ...
< Back
Next >
Finish
Cancel
Connection test
Connected (40 ms)
Server: PostgreSQL 15.19 (Debian 15.19-1.pgdg13+2)
PostgreSQL 15.19 (Debian 15.19-1.pgdg13+2) on x86_64-pc-linux-gnu, compiled by gcc (Debian 14.2.0-19) 14.2.0, 64-bit
Driver: PostgreSQL JDBC Driver 42.7.2
OK
Details >>
pg_roles 1
rolname
backend_dev
bi_dev
data_engineer
Refresh
Export data
3 row(s) fetched - 0.003s, on 2026-10-27 at 19:59:35

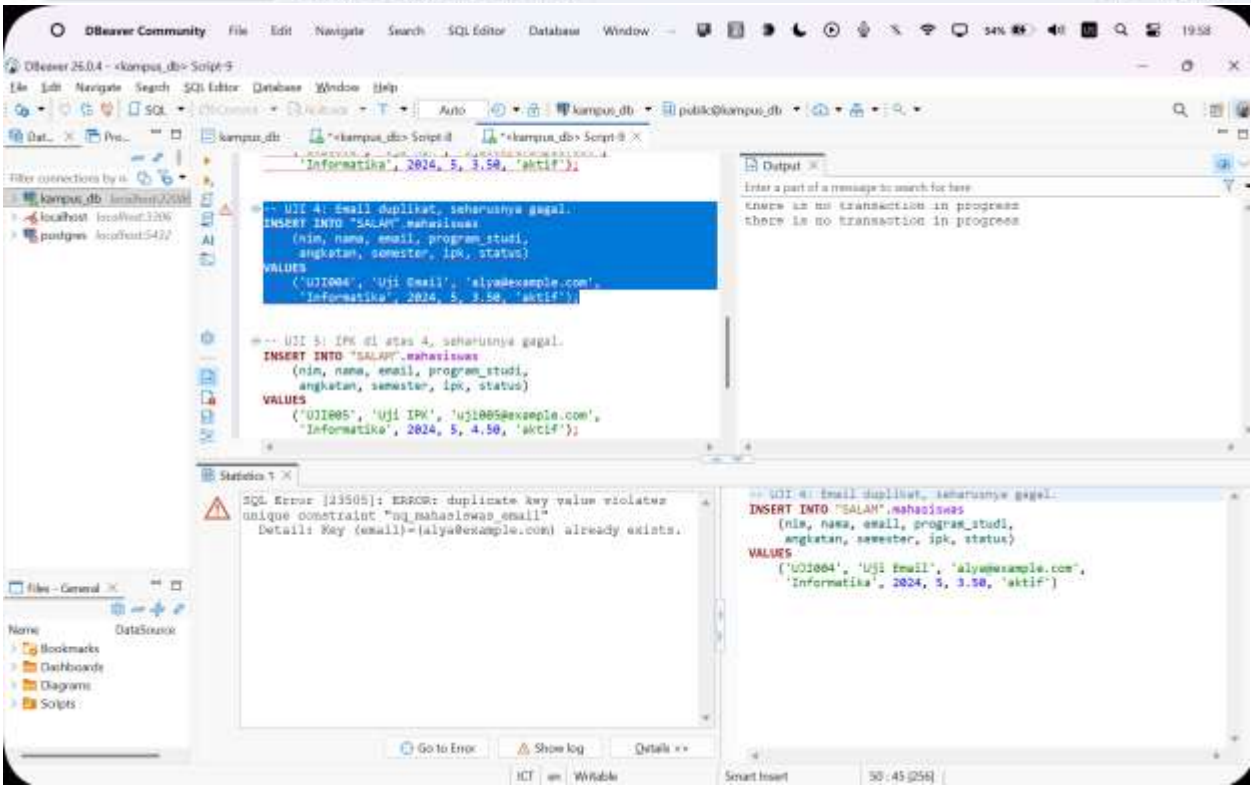

DBeaver Community
Connection Settings
PostgreSQL connection settings
Main
Advanced
Driver properties
Server
Connect by: Host URL
Host: localhost
Database: kampus_db
Authentication
Authentication: Username/password
Username: bi_dev
Password:
Test Connection ...
< Back
Finish
Cancel
Connection test
Connected (43 ms)
Server: PostgreSQL 15.19 (Debian 15.19-1.pgdg13+2)
Driver: PostgreSQL JDBC Driver 42.7.2
OK
Details >>

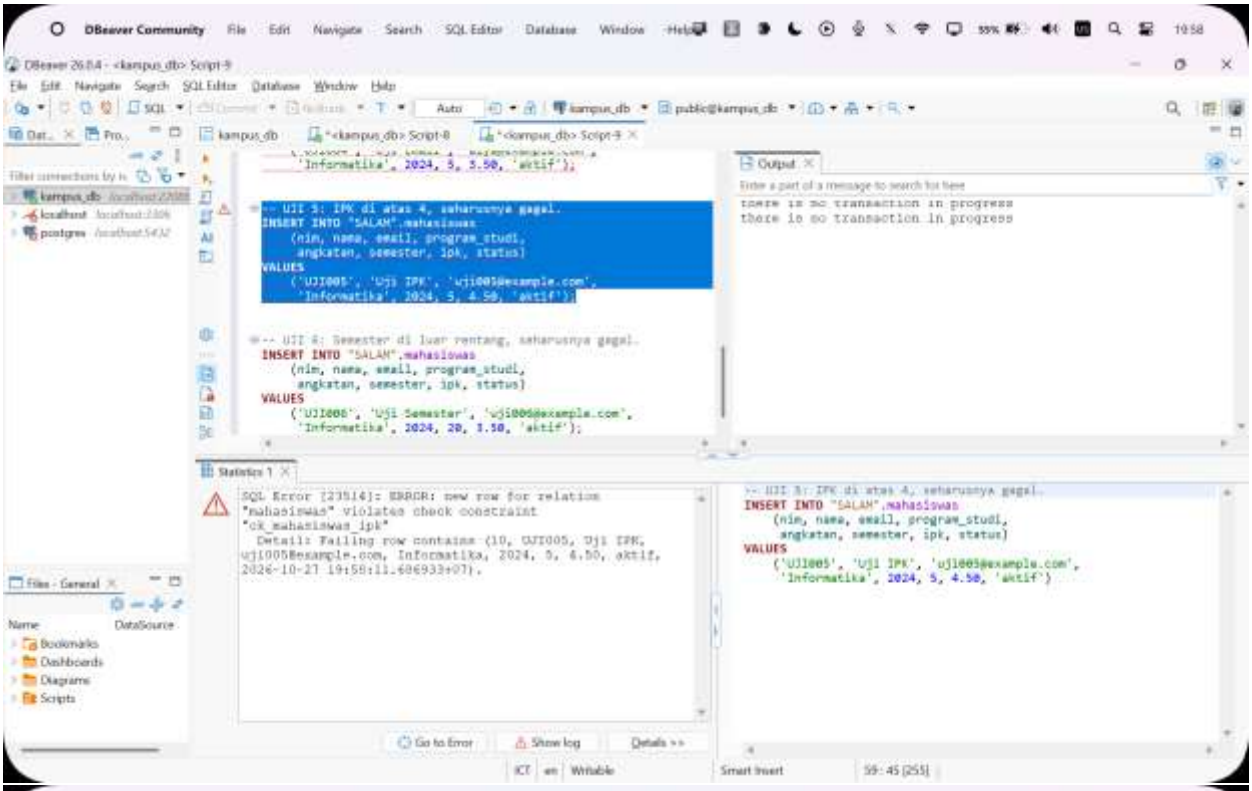

DBeaver Community
Connection Settings
PostgreSQL connection settings
Main
Advanced
Driver properties
Server
Connect by: Host URL
Host: localhost
Database: kampus_db
Authentication
Authentication: Username/password
Username: data_engineer
Password:
Test Connection ...
< Back
Finish
Cancel
Connection test
Connected (41 ms)
Server: PostgreSQL 15.19 (Debian 15.19-1.pgdg13+2)
Driver: PostgreSQL JDBC Driver 42.7.2
OK
Details >>

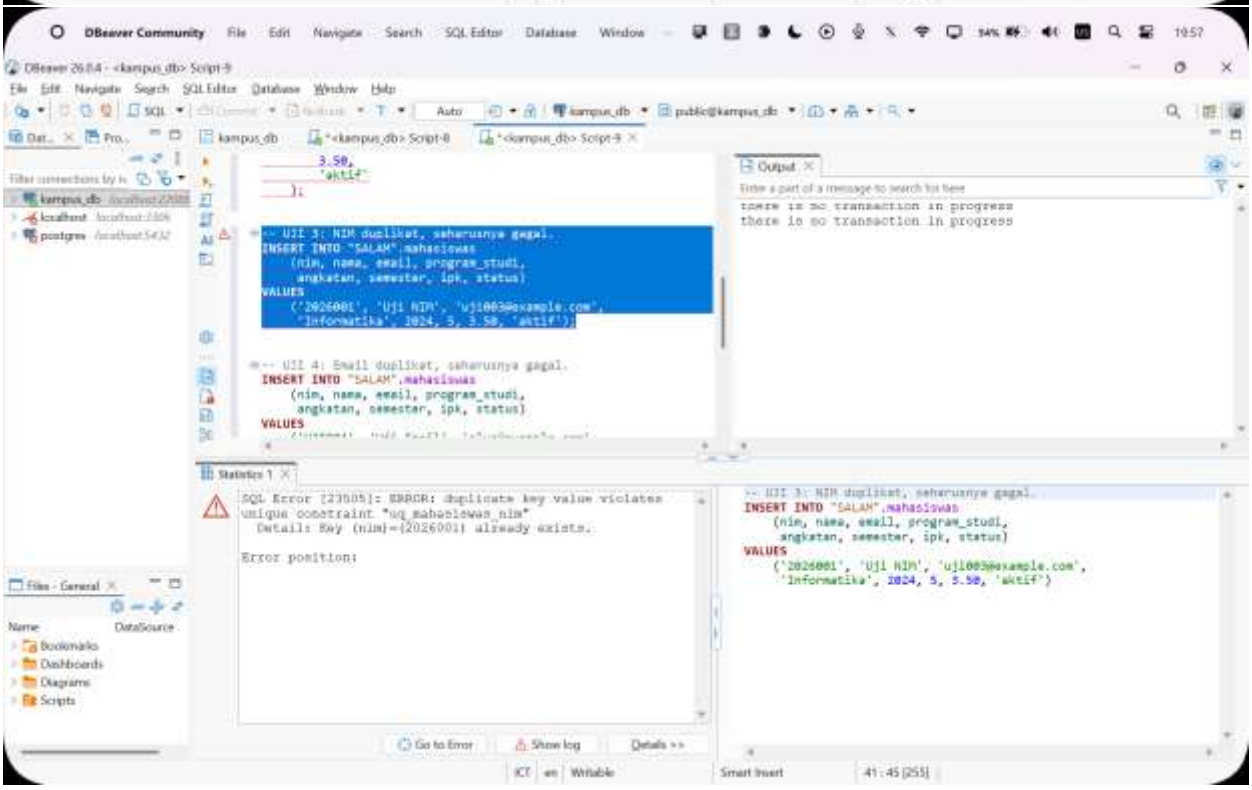

pgAdmin 4
localhost:44086/browser/
pgAdmin
File
Object
Tools
Help
admin@example.com (internal)
Servers (1)
PostgreSQL Proyek 1
Databases (2)
kampus_db
Casts
Catalogs
Event Triggers
Extensions
Foreign Data Wrappers
Languages
Publications
Schemas
Subscriptions
postgres
Login/Group Roles
Tablespaces
Connection Warning
The application has lost the database connection.
- If the connection was idle it may have been forcibly disconnected.
- The application server or database server may have been restarted.
- The user session may have timed out.
Do you want to continue and establish a new session
Cancel
Continue
Data Output
Messages
Notifications

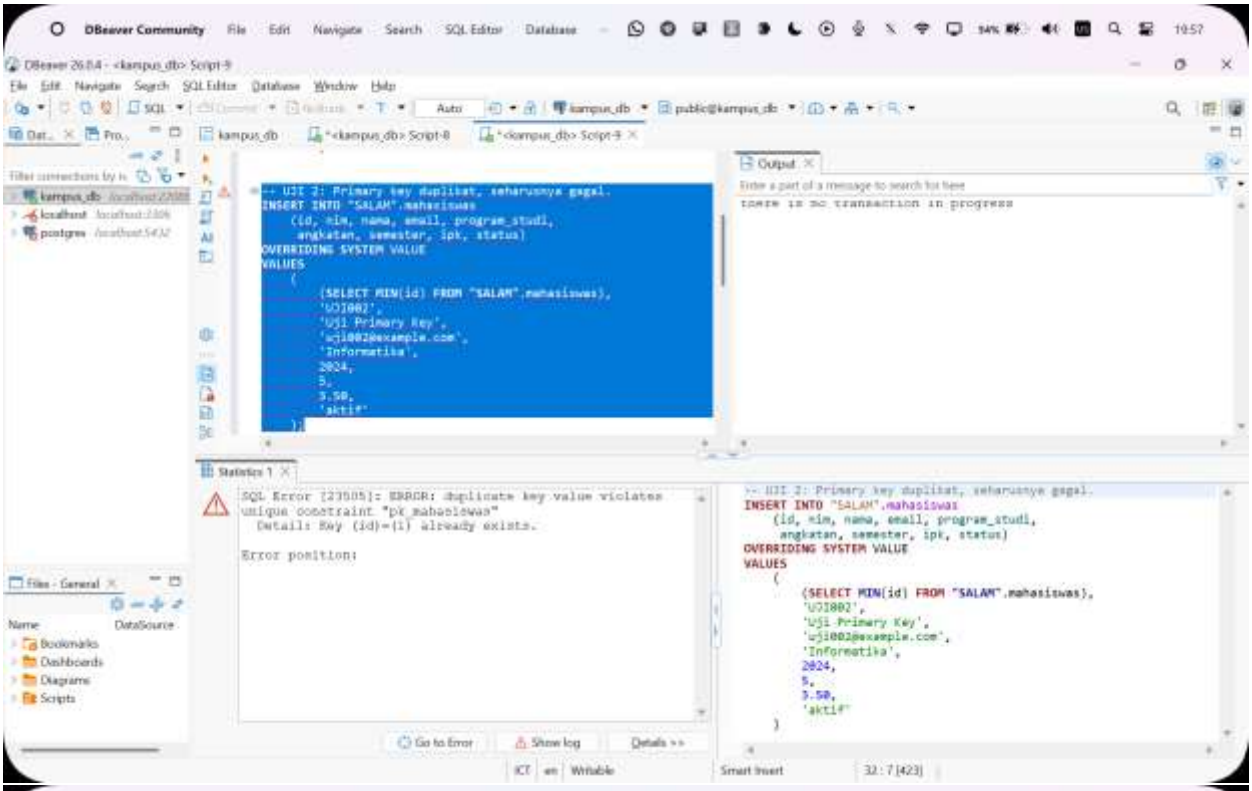

pgAdmin
File
Object
Tools
Help
admin@example.com (internal)
Dashboard
Properties
SQL
Statistics
Dependencies
Dependents
Processes
Servers (1)
PostgreSQL Proyek 1
Databases (2)
kampus_db
Connection
Connected?
Connection String
Host name/address
postgres
Port
5432
Maintenance database
kampus_db
Username
postgres
Kerberos authentication?
GSS authenticated?
GSS encrypted?
Role
Service
SSH Tunnel
Use SSH tunneling

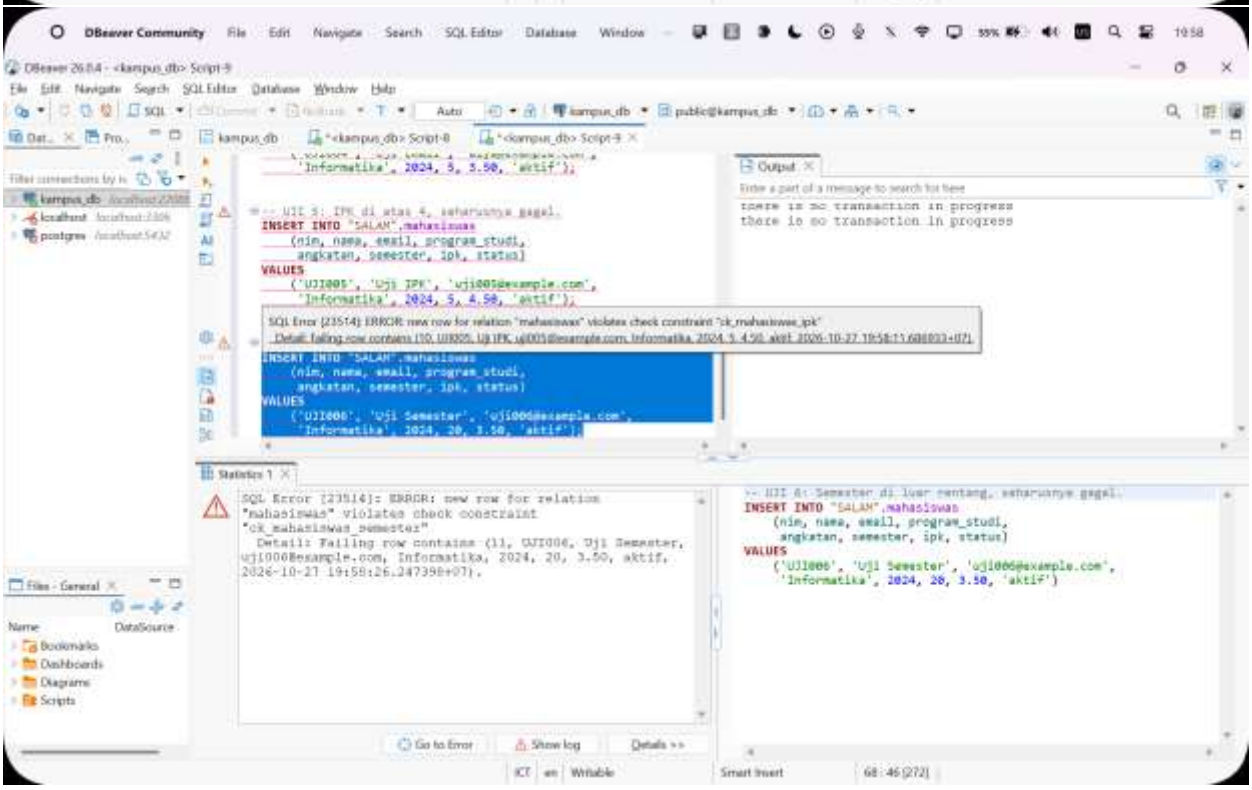

pgAdmin
File
Object
Tools
Help
admin@example.com (internal)
Object Explorer
Servers (1)
PostgreSQL Proyek 1
Databases (2)
kampus_db
Casts
Catalogs
Event Triggers
Extensions
Foreign Data Wrappers
Languages
Publications
Schemas
Subscriptions
postgres
Login/Group Roles
Tablespaces
Properties
SQL
Statistics
Dependencies
Dependents
Processes
kampus_db/postgres@PostgreSQL Proyek 1
Query
Query History
Scratch Pad
SELECT
current_database() AS database_aktif,
current_user AS pengguna,
inet_server_addr() AS alamat_server,
inet_server_port() AS port_server;
SELECT *
FROM "SALAM".mahasiswas
ORDER BY id;
Data Output
Messages
Notifications
Showing rows: 1 to 3
Page No: 1
of 1
id
nim
nama
email
program_studi
angkatan
semester
ipk
1
2026001
Alya Putri
alya@example.com
Informatika
2024
3
3.80
2
2026002
Bima Pratama
bima@example.com
Informatika
2024
3
3.20
3
2026003
Citra Lestari
citra@example.com
Sistem Informasi
2023
7
3.90
Total rows: 3
Query complete 00:00:00.127
CRLF
Ln 9, Col 13

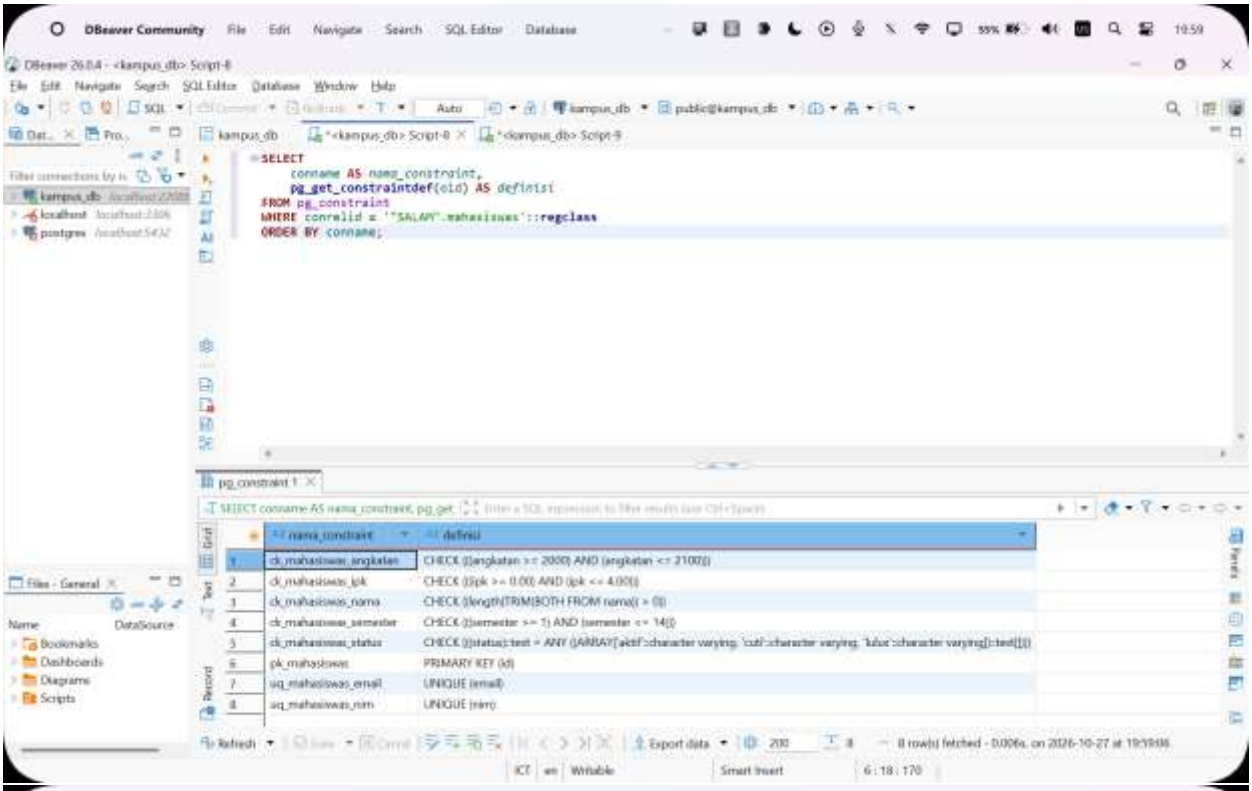

DBeaver Community
File
Edit
Navigate
Search
SQL Editor
Database
Window
Help
kampus_db
public@kampus_db
'Informatika', 2024, 5, 3.50, 'aktif');
-- UJI 4: Email duplikat, seharusnya gagal.
INSERT INTO "SALAM".mahasiswas
(nim, nama, email, program_studi,
angkatan, semester, ipk, status)
VALUES
('UJI004', 'Uji Email', 'alya@example.com',
'Informatika', 2024, 5, 3.50, 'aktif');
-- UJI 5: IPK di atas 4, seharusnya gagal.
INSERT INTO "SALAM".mahasiswas
(nim, nama, email, program_studi,
angkatan, semester, ipk, status)
VALUES
('UJI005', 'Uji IPK', 'uji005@example.com',
'Informatika', 2024, 5, 4.50, 'aktif');
Output
there is no transaction in progress
there is no transaction in progress
Statistics 1
SQL Error [23505]: ERROR: duplicate key value violates unique constraint "uq_mahasiswas_email"
Detail: Key (email)=(alya@example.com) already exists.
Go to Error
Show log
Details
Files - General
Bookmarks
Dashboards
Diagrams
Scripts

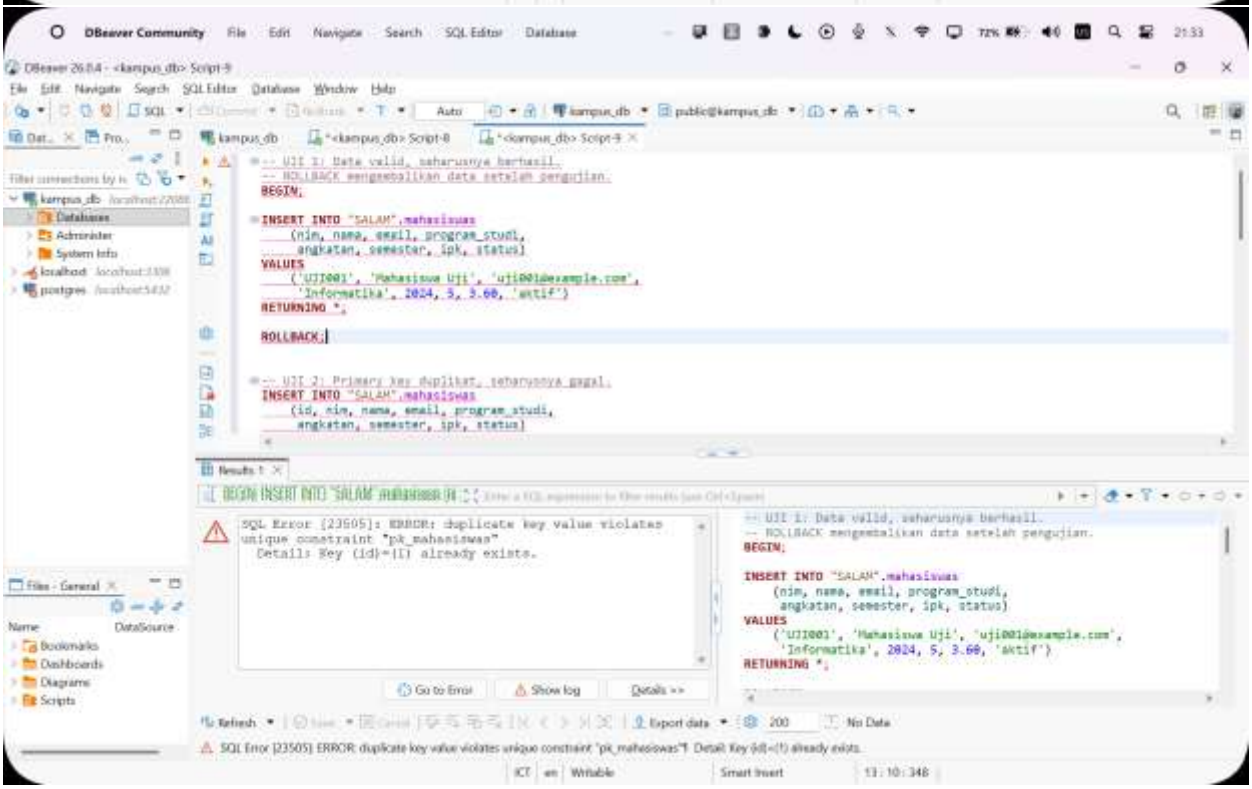

DBeaver Community
-- UJI 3: IPK di atas 4, seharusnya gagal.
INSERT INTO "SALAM".mahasiswas
(nim, nama, email, program_studi,
angkatan, semester, ipk, status)
VALUES
('UJI005', 'Uji IPK', 'uji005@example.com',
'Informatika', 2024, 5, 4.50, 'aktif');
-- UJI 4: Semester di luar rentang, seharusnya gagal.
INSERT INTO "SALAM".mahasiswas
(nim, nama, email, program_studi,
angkatan, semester, ipk, status)
VALUES
('UJI006', 'Uji Semester', 'uji006@example.com',
'Informatika', 2024, 20, 3.50, 'aktif');
there is no transaction in progress
there is no transaction in progress
SQL Error [23514]: ERROR: new row for relation
"mahasiswas" violates check constraint
"ck_mahasiswas_ipk"
Detail: Failing row contains (10, UJI005, Uji IPK,
uji005@example.com, Informatika, 2024, 5, 4.50, aktif,
2026-10-27 19:58:11.606933+07).
Go to Error
Show log
Details >>
59 : 45 [255]

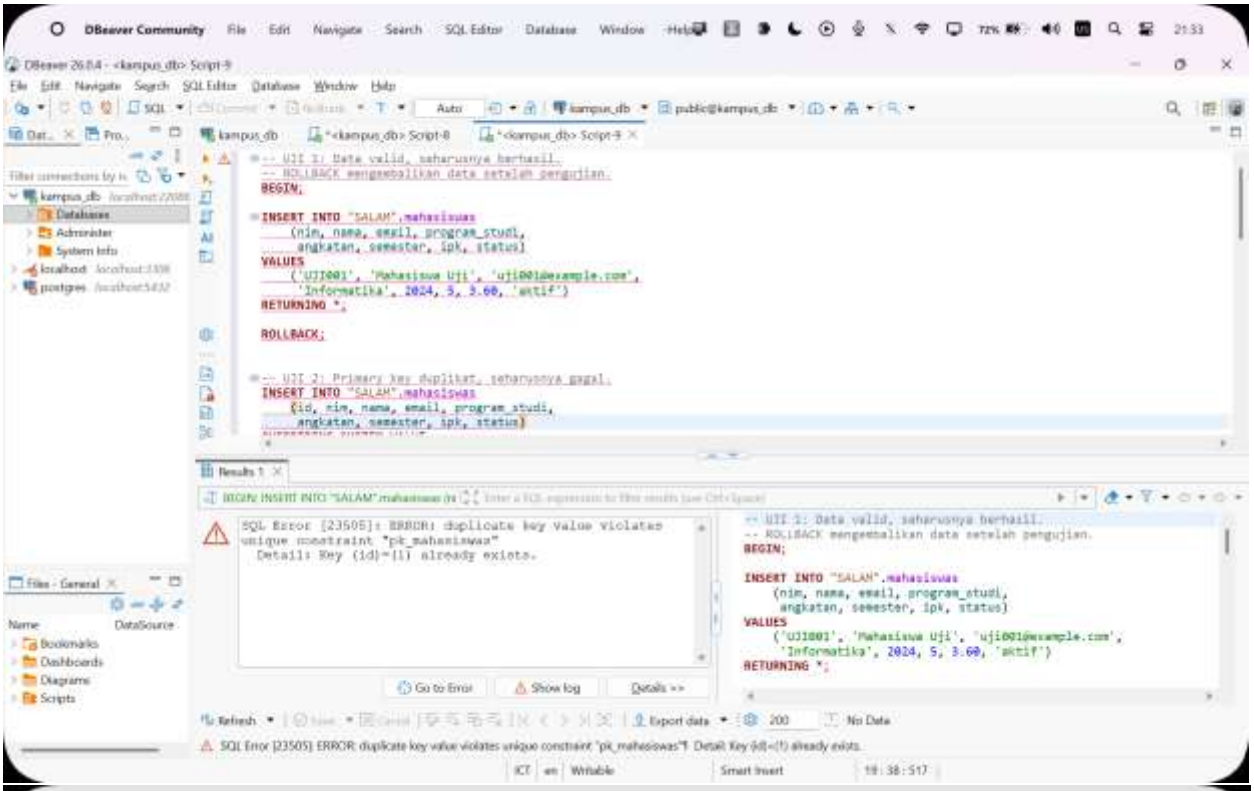

DBeaver Community
-- UJI 3: NIM duplikat, seharusnya gagal.
INSERT INTO "SALAM".mahasiswas
(nim, nama, email, program_studi,
angkatan, semester, ipk, status)
VALUES
('2026001', 'Uji NIM', 'uji003@example.com',
'Informatika', 2024, 5, 3.50, 'aktif');
-- UJI 4: Email duplikat, seharusnya gagal.
INSERT INTO "SALAM".mahasiswas
(nim, nama, email, program_studi,
angkatan, semester, ipk, status)
VALUES
there is no transaction in progress
there is no transaction in progress
SQL Error [23505]: ERROR: duplicate key value violates
unique constraint "uq_mahasiswas_nim"
Detail: Key (nim)=(2026001) already exists.
Error position:
Go to Error
Show log
Details >>
41 : 45 [255]

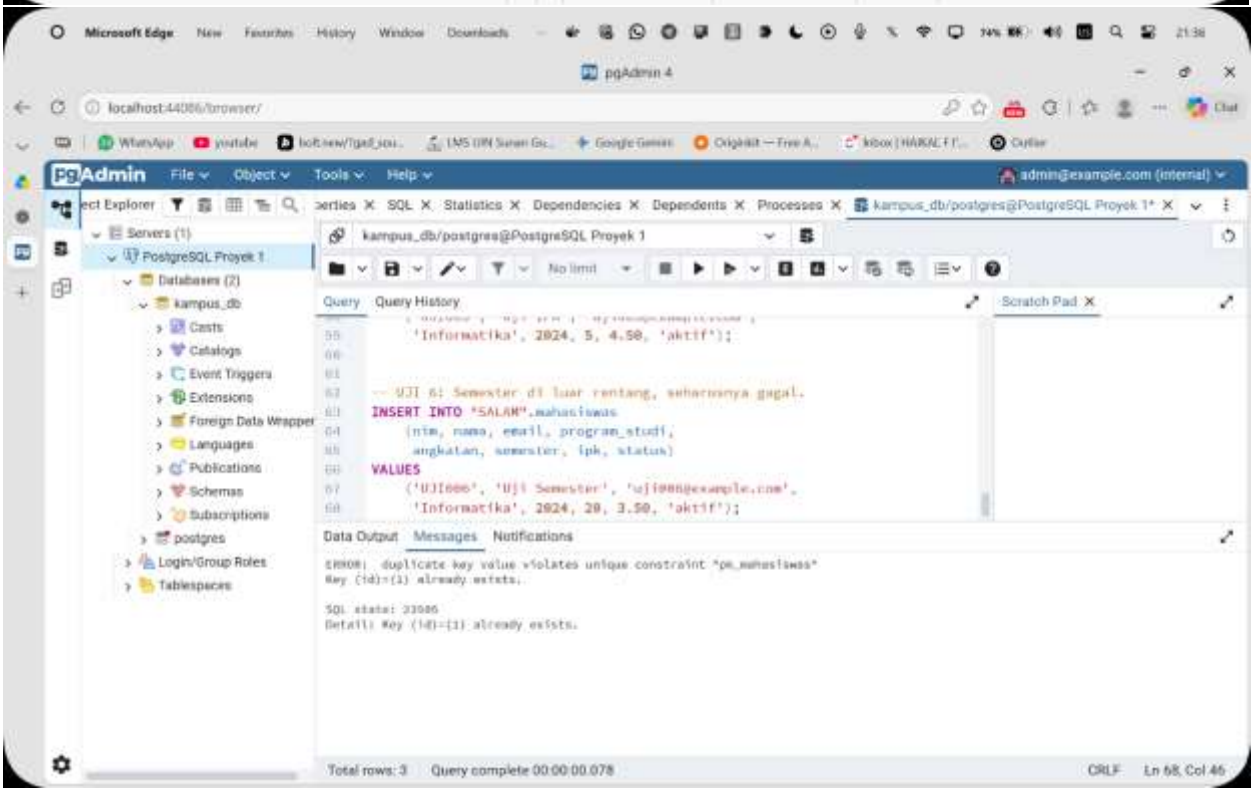

DBeaver Community
-- UJI 2: Primary key duplikat, seharusnya gagal.
INSERT INTO "SALAM".mahasiswas
(id, nim, nama, email, program_studi,
angkatan, semester, ipk, status)
OVERRIDING SYSTEM VALUE
VALUES
(
(SELECT MIN(id) FROM "SALAM".mahasiswas),
'UJI002',
'Uji Primary Key',
'uji002@example.com',
'Informatika',
2024,
5,
3.50,
'aktif'
);
there is no transaction in progress
SQL Error [23505]: ERROR: duplicate key value violates unique constraint "pk_mahasiswas"
Detail: Key (id)=(1) already exists.
Error position:
Go to Error
Show log
Details >>

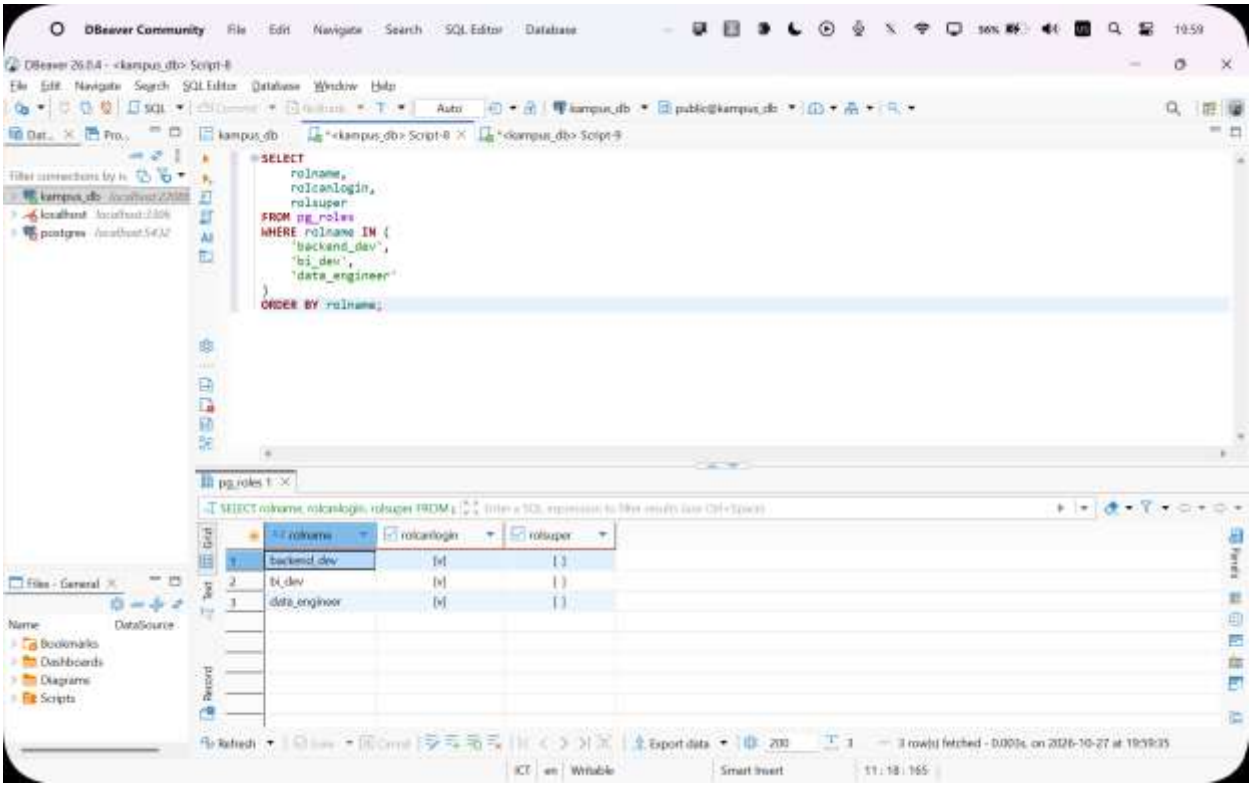

DBeaver Community
'Informatika', 2024, 5, 3.50, 'aktif');
-- UJI 3: IPK di atas 4, seharusnya gagal.
INSERT INTO "SALAM".mahasiswas
(nim, nama, email, program_studi,
angkatan, semester, ipk, status)
VALUES
('UJI005', 'Uji IPK', 'uji005@example.com',
'Informatika', 2024, 5, 4.50, 'aktif');
SQL Error [23514]: ERROR: new row for relation "mahasiswas" violates check constraint "ck_mahasiswas_ipk"
Detail: Failing row contains (10, UJI005, Uji IPK, uji005@example.com, Informatika, 2024, 5, 4.50, aktif, 2026-10-27 19:58:11.688803+07).
-- UJI 6: Semester di luar rentang, seharusnya gagal.
INSERT INTO "SALAM".mahasiswas
(nim, nama, email, program_studi,
angkatan, semester, ipk, status)
VALUES
('UJI006', 'Uji Semester', 'uji006@example.com',
'Informatika', 2024, 20, 3.50, 'aktif');
there is no transaction in progress
there is no transaction in progress
SQL Error [23514]: ERROR: new row for relation "mahasiswas" violates check constraint "ck_mahasiswas_semester"
Detail: Failing row contains (11, UJI006, Uji Semester, uji006@example.com, Informatika, 2024, 20, 3.50, aktif, 2026-10-27 19:58:26.247398+07).
Go to Error
Show log
Details >>


DBeaver Community
SELECT
conname AS nama_constraint,
pg_get_constraintdef(oid) AS definisi
FROM pg_constraint
WHERE conrelid = '"SALAM".mahasiswas'::regclass
ORDER BY conname;
nama_constraint
definisi
ck_mahasiswas_angkatan
ck_mahasiswas_ipk
ck_mahasiswas_nama
ck_mahasiswas_semester
ck_mahasiswas_status
pk_mahasiswas
uq_mahasiswas_email
uq_mahasiswas_nim


DBeaver Community
-- UJI 1: Data valid, seharusnya berhasil.
-- ROLLBACK mengembalikan data setelah pengujian.
BEGIN;
INSERT INTO "SALAM".mahasiswas
(nim, nama, email, program_studi,
angkatan, semester, ipk, status)
VALUES
('UJI001', 'Mahasiswa Uji', 'uji001@example.com',
'Informatika', 2024, 5, 3.60, 'aktif')
RETURNING *;
ROLLBACK;
-- UJI 2: Primary key duplikat, seharusnya gagal.
INSERT INTO "SALAM".mahasiswas
(id, nim, nama, email, program_studi,
angkatan, semester, ipk, status)
SQL Error [23505]: ERROR: duplicate key value violates unique constraint "pk_mahasiswas"
Detail: Key (id)=(1) already exists.


-- UJI 1: Data valid, seharusnya berhasil.
-- ROLLBACK mengembalikan data setelah pengujian.
BEGIN;
INSERT INTO "SALAM".mahasiswas
(nim, nama, email, program_studi,
angkatan, semester, ipk, status)
VALUES
('UJI001', 'Mahasiswa Uji', 'uji001@example.com',
'Informatika', 2024, 5, 3.60, 'aktif')
RETURNING *;
ROLLBACK;
-- UJI 2: Primary key duplikat, seharusnya gagal.
INSERT INTO "SALAM".mahasiswas
(id, nim, nama, email, program_studi,
angkatan, semester, ipk, status)
SQL Error [23505]: ERROR: duplicate key value violates
unique constraint "pk_mahasiswas"
Detail: Key (id)=(1) already exists.
Go to Error
Show log
Details >>
SQL Error [23505] ERROR: duplicate key value violates unique constraint "pk_mahasiswas" Detail: Key (id)=(1) already exists.


pgAdmin 4
localhost:44086/browser/
kampus_db/postgres@PostgreSQL Proyek 1
'Informatika', 2024, 5, 4.50, 'aktif');
-- UJI 6: Semester di luar rentang, seharusnya gagal.
INSERT INTO "SALAM".mahasiswas
(nim, nama, email, program_studi,
angkatan, semester, ipk, status)
VALUES
('UJI006', 'Uji Semester', 'uji006@example.com',
'Informatika', 2024, 20, 3.50, 'aktif');
Data Output
Messages
Notifications
ERROR: duplicate key value violates unique constraint "pk_mahasiswas"
Key (id)=(1) already exists.
SQL state: 23505
Detail: Key (id)=(1) already exists.
Total rows: 3
Query complete 00:00:00.078
Ln 68, Col 46


DBeaver Community
File Edit Navigate Search SQL Editor Database
DBeaver 26.0.4 - <kampus_db> Script-8
File Edit Navigate Search SQL Editor Database Window Help
kampus_db
"<kampus_db> Script-8
"<kampus_db> Script-9
SELECT
rolname,
rolcanlogin,
rolsuper
FROM pg_roles
WHERE rolname IN (
'backend_dev',
'bi_dev',
'data_engineer'
)
ORDER BY rolname;
pg_roles 1
rolname
rolcanlogin
rolsuper
1 backend_dev [x] [ ]
2 bi_dev [x] [ ]
3 data_engineer [x] [ ]
Files - General
Name
DataSource
Bookmarks
Dashboards
Diagrams
Scripts
Refresh
Export data
3 row(s) fetched - 0.003s, on 2026-10-27 at 19:59:35
Smart Insert